

kayfabe v3 — the architecture, the machine as it runs, and every constant we touch
A working document, assembled 2026-09-20 (w821). Written to be argued with.

branch w749-fable-legb

Contents

Read this first	4
What changed since the v2 bundle (w818)	4
The markers, used consistently throughout	4
Part 1 — The v3 architecture (the proposal)	5
0. □□□□ THE SEVEN AXES — every constant in this design varies along at least one	5
0.1 The axis band for (Dg, Dh) is a DIAGONAL, not a product	5
0.2 □□□ The axes are not evenly distributed across the four planes	6
0.3 ∅ What this forbids, concretely	6
1. The shape	7
2. The vCPU trap path	7
2.1 □□□□ The privilege boundary INSIDE the guest is the organising idea	7
2.2 The doorbell plane — a bit table, and the hint ring is DELETED	8
2.3 The privileged plane — one ordered ring, and per-vCPU rings were WRONG	9
2.4 The wakeup protocol	9
2.5 The doorbell page is READ-ONLY, not emulated	10
3. Memory	11
4. Addresses	11
4.1 Mirror, never adopt — and the review’s reasons are stronger than mine	11
4.2 The diff	11
4.3 A VA space is the OBJECT. The PDB is an attribute of it, per GPU.	11
4.4 Where the root actually comes from — it is ours to write	12
5. Completions and interrupts	12
6. The scrub	13
7. Portability — derive, do not hardcode	13
8. What is deleted	13
9. Rewrite, not refactor	14
10. Still open	14
Part 2 — The surface, at constant level	15
0. The four planes, and why they are different shapes	15
1. GSP RPC — the plane we answer as the firmware	15
1.1 What this plane is, for a reader who does not know NVIDIA’s constants	15
1.2 The structures on the wire	15
1.3 The functions	16
1.4 □ The reply is always built from local state	19
.	19
3. BAR0 MMIO — reads from DRAM, writes to the vCPU	19
3.1 The asymmetry, and why it exists	19
3.2 The shadow — two structures, and they are not the same thing	19
3.3 The trapped-write register table	20
3.4 □□□ WPR2 is the worked example of two axes crossing in one register	22
3.5 Whole apertures, not registers	22
4. Channels and pushbuffers — the plane unprivileged guest code drives	23
4.1 What a channel is, for a reader who does not know NVIDIA’s constants	23
4.2 The three kinds — and an honest statement of what is built	23
4.3 The GPFIFO entry — 8 bytes	24
4.4 The method header — one dword	24
4.5 Every pushbuffer method we recognise	25
4.6 What happens when each kind’s doorbell rings	26

Part 3 — The machine, as it is	28
0. How to read this	28
2. Processes and threads	29
2.1 The processes	29
2.2 The sandbox — two stages, forced order	30
2.3 The threads	30
3. The vCPU trap path, step by step	31
3.1 How a guest access reaches Rust	31
3.2 A DOORBELL write (BAR0 0x00BB0090) — the hot path	31
3.3 An MMU_INVALIDATE write (BAR0 0x00B830B0) — the one the guest spin-polls	31
3.4 Back-pressure — the rule that makes deferral safe	32
4. Locks, and the blocking rules	32
4.1 The ranked order	32
4.2 The blocking rules — what is actually enforced	32
4.3 Ø The gate that is RED right now	33
4.4 Where we block on purpose	33
5. Data structures	33
5.1 The object graph — identity vs label	33
5.2 The runtime model	34
5.3 The address table — what v2 deletes	34
5.4 Crate layering	34
5.5 The isolate boundary — two layers, not one	34
6. Installing a mapping, and observing a completion	35
6.1 The mapping chain, end to end	35
6.2 The single store	35
6.3 Completions — two mechanisms with opposite writers	35
6.4 Interrupts — what gets announced	36
7. The proposal, and the arguments against it	36
P1. Mirror, not adopt	36
P2. No address table; the host page tables are the model	36
P3. Lazy everywhere; strict only at recycle — and recycle is the guest's job	36
P4. No RM verb on a vCPU, structurally	36
P5. Deletion order	36
P6. What must NOT be deleted	36
8. Open problems — named, not designed away	37
Part 4 — The surface we present (v2 plan-level)	38
1. GSP RPCs	38
1.1 The object-model RPCs — KEEP, unchanged	38
1.2 UPDATE_BAR_PDE (0x46) — KEEP, but it is the BAR2 root	38
1.3 SET_GUEST_SYSTEM_INFO (0x01) / _EXT (0x40) — KEEP	38
1.4 GET_GSP_STATIC_INFO (0x41) — KEEP, forged by construction	38
1.5 The inert set — KEEP as inert, they are load-bearing	38
1.6 Events we post — KEEP	38
1.7 Everything else — refused by name, and that is the design	39
2. RM control commands	39
2.1 Page-directory controls — [DELETE the concept, keep the decode]	39
2.2 GPU_PROMOTE_CTX (0x2080012b) — KEEP	39
2.3 Channel controls served by ObjectPolicy — KEEP	39
2.4 The 47 forged tables — KEEP forged, but they are a per-die liability	39
2.5 Input-only controls — KEEP	39
2.6 Denied by capability — KEEP DENIED	40
3. MMIO registers	40
3.1 Boot constants — KEEP	40
3.2 The GSP boot FSM — KEEP	40
3.3 Queue doorbells and the interrupt tree — KEEP	40
3.4 The MMU invalidate trio — [DELETE most of what it triggers]	40
3.5 The framebuffer windows — [DELETE]	40
3.6 The doorbell — KEEP, it is the hot path	40
4. Channel / pushbuffer operations	41
4.1 EMULATED — we implement the effect	41
4.2 TRANSLATED — operands rewritten, the GPU moves the bytes	41
4.3 REFUSED — the fault tripwire	41
4.4 PASSED THROUGH	41

5. □ Where this inventory is knowingly incomplete	41
Part 5 — The scrub question (open owner ruling)	43
What is measured	43
Why this is a decision and not a bug	43
The argument each way	43
If the ruling is “\$46 wins”, the work and its gate	43
What must not be read into this page	44

Read this first

This bundle is five documents, in the order they should be read:

1. **The v3 architecture** — the design I propose, rewritten after the owner’s review and after an adversarial review of the worker protocol. Short, and the part to attack first.
2. **The surface, at constant level** — *new in this edition*. Every GSP RPC, RM object class, RM control command, BAR0 register and pushbuffer method we touch, **by its NVIDIA constant name and hex value**, each with a plain-language sentence saying what it *is* for a reader who does not know NVIDIA’s constants, and what we do with it.
3. **The machine, as it is** — what the code actually does today, from a file:line survey. Every place the code contradicts the proposal is marked.
4. **The surface we present (v2)** — the earlier plan-level inventory: KEEP / DELETE per item. Kept because the *plans* in it are still the plans; document 2 supersedes its *data*.
5. **The scrub question** — one open owner ruling, isolated because it is a live security defect and it is the smallest change on the list.

What changed since the v2 bundle (w818)

- **☐☐☐ A second privilege boundary.** [owner] The doorbell is adversarial **to the guest’s own root**, because unprivileged guest userspace writes it. This is not hardening bolted on — it restructures the entire vCPU trap path. v3 §2.
- **∅ The doorbell hint queue and every FULL_REFRESH-shaped fallback are DELETED**, replaced by a 64 KiB bit table. Deleted from the *proposal*, before any code was written.
- **∅ Per-vCPU register rings are refuted** — they preserve per-vCPU order where the device needs the guest’s cross-vCPU order.
- **☐ The worker wakeup protocol is fully specified**, with its bit widths, its layout, its memory orderings and the interleavings that break it.
- **☐ This document, part 2** — the constant-level reference the v2 bundle did not have.

The markers, used consistently throughout

marker	meaning
[MEASURED]	a number from a real boot, with the run that produced it
[CODE]	read out of the tree, with file:line
[ogkm]	read out of NVIDIA’s open kernel modules, with file:line
[PROPOSE]	a design opinion. These are the arguable ones.
[UNVERIFIED]	the survey could not establish it. Deliberately not filled in with a guess.
∅	a refusal, a deletion, or a thing that is <i>not</i> true
☐	a load-bearing fact
☐	a trap, or a claim whose evidence is weaker than it looks

☐ **Nothing in this bundle is filled in from memory of NVIDIA’s headers.** Every constant name, number and structure layout is cited to a file and line in either this tree or the open kernel modules under `research_clones/ogkm`. Where a survey could not establish something, it says **[UNVERIFIED]** rather than guessing — that convention is load-bearing, because this project’s most expensive recurring failure is a plausible value that was never measured.

Part 1 — The v3 architecture (the proposal)

STATUS: LIVE, 2026-09-20 (w820). **Supersedes** `THE_ARCHITECTURE_v2.md`, which this document keeps only as the record of what was proposed before the review corrected it.

∅ **w820 rewrote §2 entirely** around a second privilege boundary the owner named: the one **inside the guest**, between unprivileged guest userspace and guest root. The doorbell hint queue and its “inspect all doorbells” fallback are **deleted**, not fixed; per-vCPU register rings are **refuted**; the worker wakeup protocol is specified with its orderings. §§1, 8 and 10 were updated to match. Everything else is unchanged from w819.

∅ Written from the owner’s point-by-point review. Where the review corrected me, the correction is marked **[CORRECTED]** with what I had wrong — those are the load-bearing edits, not footnotes.

0. □□□□ THE SEVEN AXES — every constant in this design varies along at least one

[NEW, w821] [owner, 2026-09-20] listed the axes this product must span. They are the reason almost nothing in Part 2 may be a literal in our source, and they are **prior to** every other section: a design that is correct on one point of this lattice and cannot be extended to the rest is a retrofit waiting to happen.

tag	axis	posture	how variation is absorbed
K	guest kernel version	∅ all major versions	never our concern directly — but it sets what the guest driver is <i>built against</i> , so it moves Dg’s struct layouts
Dg	guest driver version	∅ all major versions	generated from that version’s headers; a per-version profile entry , never if version ==
Dh	host driver version	∅ all major versions, decoupled from Dg	we author every host call; our host-verb signatures do not accept a guest flag word
A	GPU architecture	□ Turing and newer	a format family descriptor (four families span Turing→Blackwell)
die	GPU die within an arch	∅ any die	derived per die, maintained per family ; a new die is a descriptor, not a code path
V	VMM	● the one axis where a version floor is legitimate — QEMU, Cloud Hypervisor	the core is VMM-agnostic; the VMM shim is the only place that knows
OS	guest OS	Linux and Windows	□ the axis with the least coverage; everything measured in this campaign used a Linux guest

∅ **Dg and Dh are TWO axes, not one.** The operator chooses the guest driver; we do not. A design that assumes they match is a defect. □ This is the structural reason for `author_host_flags_never_forward_them`: forwarding a guest-supplied flag word into a host RM call is not merely risky, it is **an implicit assertion that the two versions agree**.

∅ **Host OS is Linux only** — the one scope relaxation, taken deliberately.

0.1 The axis band for (Dg, Dh) is a DIAGONAL, not a product

[owner ruling, 2026-08-09] We owe NVIDIA vGPU’s own interoperability policy and no more: exact match; same major branch different minor; and **n−1** (a newer host supports the previous major or LTS branch). ⇒ A bounded window, not every pair.

□ What that **obliges us to build**: we must know *both* versions and be able to say “*this pair is outside the band*” — **refused by name**. ∅ A compatibility policy that cannot state an out-of-band pair is not a policy, it is a hope, and it decays into the 0x56 is the forgiven status trap where a wrong configuration simply runs on.

0.2 □□□ The axes are not evenly distributed across the four planes

This is the useful part, and the surveys in Part 2 measured it rather than assuming it:

plane	dominant axis	evidence
GSP RPC function numbers	□ stable across Dg in the measured window	the NV_VGPU_MSG_FUNCTION_* enum is byte-identical between 580.159.04 and 610.43.02 for every id we use; 610 only <i>appends</i> (rpc_global_enums.h)
GSP RPC message framing	∅ Dg, and it BREAKS	the per-element header is 48 bytes at 580 (authTag/aad/checkSum@32/seqNum@36/elemCount@40) and 16 bytes at 610 (mctpHeader/nvdmHeader/checkSum@8/seqNum@12) — message_queue_priv.h:43-51 VS :52-67
RM control command numbers	Dg (additive), struct layouts break	FINN-generated; the <i>number</i> is stable, the <i>parameter struct</i> is versioned
BAR0 register offsets	A and die	NV_PGSP_QUEUE_HEAD = 0x110c00 from ampere/ga102/dev_gsp.h; the doorbell is 0x30090 on Turing, Ampere and Blackwell
Doorbell token encoding	∅ die , within one arch	NV_CTRL_VF_DOORBELL field widths are per-arch swref, and GB202 sets bit 30 where Ampere does not
Channel / engine class ids	A	VOLTA_CHANNEL_GPFIFO_A 0xC36F → AMPERE_ 0xC56F → HOPPER_ 0xC86F → BLACKWELL_ 0xC96F
Pushbuffer method encoding	□ remarkably stable	NVC56F_DMA_SEC_OP/METHOD_ADDRESS/_SUBCHANNEL/_COUNT is bit-identical to NVC36F_*; the GPFIFO entry differs only in PRIV (dropped at C56F) and INVAL_SCOPE (added)
Page-table format	∅ A , four families	Turing binds kgmmuFmtInitLevels_GP10X; our one built format is _GA10X

⇒ □□□ **The two planes that break hardest are the two the guest drives most**: RPC framing (Dg) and page-table format (A). Both are already modelled as **descriptors** rather than code paths — ElementLayout { hdr_size, checksum_off, seqnum_off, elem_count_off, TransportHdr } and GmmuFmt — and that is the shape every other axis-carrying fact should take.

□ **The 580/610 element header is the worked example to argue from**. It is a structure whose *size and every field offset* changed between two supported guest driver versions, it was absorbed as data rather than as a branch, and it proves the discipline is implementable rather than aspirational.

0.3 ∅ What this forbids, concretely

1. ∅ **No literal in a comparison**. derive_per_die_maintain_per_family. Nothing may test a raw offset, class id or command number against a hardcoded constant.
2. ∅ **No C parsed with regex or sed**. Generate from the headers with a real parser front end; [owner] “*That will age better.*”
3. ∅ **No capture-derived table treated as truth**. A table captured from one driver on one die **expires as a vendor regression** — and this tree measured that 11 of 56 captured rows were empty and *every empty one checked against hardware was contradicted*.
4. ∅ **No guest flag word forwarded to a host call**. It is a Dg↔Dh coupling in disguise.
5. ∅ **No “Linux-shaped” assumption left unmarked**. Until a Windows guest runs, every guest-side inference in this document is **os-limited**, and Part 2 marks the ones that are.

□ **Where this document is weakest on the axes**, stated plainly so it is not discovered later: the constant-level tables in Part 2 were surveyed against **Ampere GA10x with a Linux guest**. The axis column in each table says what *should* vary; it does **not** certify that each entry has been exercised on a second point of that axis.

1. The shape

One process. The VMM (QEMU, or later any hypervisor) with the kayfabe core linked in. No isolate children, no scratchpad process, no IPC, no sandbox plane of our own.

Two kinds of thread besides the vCPUs:

thread	count	job
worker	N, capped at 255	everything a trap deferred: managed doorbells, GSP RPCs, TLB invalidates
VA manager	1	□ <i>new</i> . One synchronous thread doing all mmap/munmap — page-table diffs, BAR windows, channel mappings. It replaces the scratchpad isolate.

□ **[w820]** The worker cap is **structural, asserted at startup** — not a configuration convention. `workers_polling` is 8 bits of `worker_vcpu_poll` (§2.4) and a wrap would read **0 while workers are polling**, which is the lost wakeup the protocol exists to prevent.

□ The VA manager exists because mapping is the one activity that must be serialized and must never happen on a vCPU. Giving it a thread rather than a lock makes “who may map” a structural answer instead of a rule.

⊗ **Sandboxing is not ours.** Sandbox the whole hypervisor process — a container, or the hypervisor’s own jail. [owner, 2026-09-20] *“most security bugs didn’t even come from the isolates... the entire isolate plane is a new plane that needs sharing fds and handles and stuff back and forth... It by itself is a source of many bugs.”* □ This session supplied the proof: `barmirror.rs:1988` does a **cross-process IPC round trip on a vCPU while holding a lock** — a defect that cannot exist without the isolate plane. Add the 1.62 s spawn stall, `Wedge` (a failure mode that exists only because there is a socket), the fd passing, cancel-by-signal and the foreign-handle gate, and the plane’s bug surface exceeds what it protects.

⊗ **Proc stops existing. [CORRECTED]** A process is not a GPU concept. v3 speaks only in **channels, VA spaces** and **RM objects**. With isolates gone the whole per-proc container, its arena, its pending-release queue and the dup-connected-component analysis go with it.

2. The vCPU trap path

Only **BAR0**, only **writes**. BAR1/BAR2 are never trapped — ensuring that is a requirement, not an optimisation. The entry point is one function, `nvkvm_trap_bar0_write`.

2.1 □□□□ The privilege boundary INSIDE the guest is the organising idea

[NEW, w820] [owner, 2026-09-20] *“the doorbell is adversarial also to the guest itself, since unprivileged guest userspace writes to it (that’s below guest kernel) ... It’s not about the guest corrupting itself from the kernel/root in the guest, it’s about a sandboxed/container process in the guest corrupting the root in the guest.”*

This is not hardening bolted onto the design. It **splits the trap path in two**, and every property each half needs is derived from which side of the line it sits on:

plane	who can write it	what it may therefore be
doorbell (usermode window)	unprivileged guest userspace	fixed-size, idempotent, non-degradable, needs no identity = a bit table, no queue
privileged registers (everything else in BAR0)	guest root only (RM)	an ordered ring that must not overflow — a guest root that overflows it harms only itself

⇒ The test for any future trapped page is one question: **can unprivileged guest userspace write it?** If yes it gets no queue, no degradation mode and no shared structure. If that is not achievable for some page, we do not map it.

Guest-root-corrupts-guest-root is inside the guest’s own trust model and not ours to defend. Guest-container-corrupts-guest-root is the value proposition.

Why no check at the trap can fix this — three facts from ogkm

1. **The doorbell page is ONE page per GPU, aliased to every client.** `usrmodeConstruct_IMPL` takes `pMemDesc = pKernelFifo->pRegVF (usermode_api.c:47)` — a single per-GPU `ADDR_REGMEM memdesc (kernel_fifo_gv100.c:370)`, 64 KiB (`NVC361_NV_USERMODE_SIZE`), doorbell at `+0x090`. `usrmodeCanCopy` is `NV_TRUE`, so it is **DUP’d, not copied**. There is **no per-client and no per-channel doorbell page on Ampere**. Hopper adds `bPriv`, gated on `RS_PRIV_LEVEL_KERNEL`, but that selects `BAR1-vs-BAR0 mapping`, not which channels you may ring. □ RM waives its own privilege check on this window **by name**: `memdescSetFlag(pRegVF, MEMDESC_FLAGS_SKIP_REGMEM_PRIV_CHECK, NV_TRUE) (:374)`. Handing a `BAR0` register window to unprivileged userspace is deliberate, documented NVIDIA behaviour.
2. **The token carries no capability.** `kfifoGenerateWorkSubmitTokenHal_GA100 (kernel_fifo_ga100.c:224)` is a plain concatenation of `runlistId` and `chId`. RM’s comment “*Caller cannot make assumption about this handle*” is a request, not an enforcement: the space is small enough to enumerate by counting.
3. **The trap carries no attributable identity.** We get a vCPU, a GPA and 32 bits the attacker chose. Nothing says *who* rang, and the guest kernel’s own `ioremap` of `BAR0` and userspace’s `mmap` of the usermode window resolve to the **same GPA** — KVM memslots are keyed by guest physical address, so the trap **cannot distinguish kernel from userspace**. The ring cannot be authorized at the trap. Not with more checks. Ever.

Why hardware survives this and a naive port does not On hardware a ring means “*channel (runlist, chid) may have work*”, and the PBDMA then reads that channel’s **USERD GP_PUT** — memory only the owner can write. A hostile ring buys a redundant fetch of work the victim already published. The attacker contributes **zero data**, only a timing hint. NVIDIA evidently accepted that.

Our handler does not re-read a pointer. For a **translated** channel it parses and translates the victim’s pushbuffer; for an **emulated** channel it executes a function on the victim’s behalf. That promotes the attacker’s timing hint into a control input over victim-state processing, and it converts two things that read as engineering into security defects:

- **○ Concurrent double-take becomes a triggerable primitive.** Ring root’s token in a tight loop and get two workers walking root’s GPFIFO concurrently against one cursor: root’s work submitted twice, or its cursor walked backwards into a PBDMA ring-wrap. The four-state token word in §2.4 is therefore **the boundary**, not a latency nicety.
- **○ A degradation path becomes an amplifier.** Any “queue full ⇒ scan everything” fallback lets an unprivileged process force the VMM to walk **everyone’s** channels on the attacker’s schedule.

□ **The tension with the owner’s literal instruction, stated rather than resolved silently** [owner] “*I would not register write traps in the unprivileged bar0 page.*” The strongest form of that — **map the page writable straight through to hardware, never trap it** — is not reachable today, for the reason in fact 3: guest RM rings the *same physical register* from the kernel, so not trapping it also gives up intercepting **kernel** channel doorbells. The scrub is one of those (§6), and it is the live cross-client leak. So v3 takes the implementable reading: **trap, but let nothing behind the trap be shared, growable or degradable**.

□ The literal form becomes reachable the day **every channel rung through that page is passthrough**. That is a real future state, not a fiction, and it is the reason §2.2 is built as a table rather than a queue: when that day comes, the doorbell trap is deleted rather than rewritten.

2.2 The doorbell plane — a bit table, and the hint ring is DELETED

[CORRECTED, w820] The w819 text here said “*push the token on the queue, set this channel’s rung bit; queue full ⇒ set ‘inspect all doorbells’ flag.*” That fallback is exactly the amplifier above, and the queue is exactly the exhaustible shared structure. **Both are deleted**. The table alone carries the plane.

```
nvkvm_trap_bar0_write(off, val):
```

```
if off == DOORBELL (0x00BB0090):           # VF aperture base + NV_VIRTUAL_FUNCTION_DOORBELL 0x30090
    tok = val & TOKEN_MASK                  # mask to the two decoded fields – see below
    w   = table[tok >> 6]
    if PASSTHROUGH(tok):
        write the host doorbell INLINE. No queue, no wake, no lock. Return.
    else:
        # MANAGED (translated or emulated), or UNKNOWN
        fetch_or(table[tok >> 6], RUNG_BIT(tok))    # AcqRel
```

```

        fetch_or(summary[tok >> 15], 1 << ...)    # Release
        bump work_seq; maybe write(eventfd)       # $2.4
        Return.

    else:
        stamp and publish onto the PRIVILEGED ring ($2.3). Return.

```

⇒ **A vCPU now takes NO lock at all on the doorbell path.** The w819 text said “one lock, the queue’s”; with the queue gone there is none.

Sizing — the number, from the published register, not from the legal space. `ampere/ga100/dev_ctrl.h:26:`

```

NV_CTRL_VF_DOORBELL_VECTOR      11:0   → 12 bits → 4096 channels
NV_CTRL_VF_DOORBELL_RUNLIST_ID  22:16  → 7 bits  → 128 runlists

```

19 decoded bits ⇒ $2^{19} = 524\,288$ tokens ⇒ **a 64 KiB bit table**, plus a one-bit-per-u64 summary layer = **a 1 KiB hot index** over it. Bits 15:12 and 31:23 are undecoded, so `TOKEN_MASK` keeps only the two fields.

□ Masking rather than validating is deliberate twice over: it is what the hardware does with undecoded bits, **and** it removes an error path — no refusal, no counter, nothing to misread. Sizing to what the register can *express* rather than to what is *legal* is this tree’s own lesson, `[a_bound_on_reads_is_not_a_bound_on_emits]`: the attacker writes the register, not the legal space.

□ Scan cost is **arithmetic, not yet measured**: 1 KiB of summary is ~16 cachelines; the loaded case streams 8192 u64 compares over 64 KiB, skipping zero words. Sub-microsecond warm, low single-digit μs cold. **That is cheaper than maintaining a ring** — which is why the ring is deleted rather than repaired.

⇒ What a hostile guest process can now buy is: one `fetch_or` on a bit in a fixed table that was going to be scanned anyway, on a **per-token cacheline** so there is not even a contention point. **Exactly hardware’s cost profile: ring anything, pay a redundant look.**

□ Per-die derivation, not a copied constant: those widths come from the Ampere-family `ga100` swref header. Per \$7 they must be **generated from the headers**, and the table sized from the generated widths — `tu102` agrees today, `gb202/gb100` put the doorbell at the same `0x30090` but their field widths must be read, not assumed.

2.3 The privileged plane — one ordered ring, and per-vCPU rings were WRONG

[CORRECTED, w820 — this defect was in my proposal, found by an adversarial review.] I had proposed per-vCPU register rings with a single-bit owner claim. Per-vCPU rings preserve only **per-vCPU** order, and the order the device needs is the one the **guest** established, which spans vCPUs:

```

T1 on vCPU0, holding an RM lock:  write PDB_LO, PDB_HI → ring0
T1 releases the lock
T2 on vCPU1, takes the lock:      write TRIGGER          → ring1
Worker B drains ring1 first → TRIGGER fires against the OLD PDB

```

On hardware this cannot happen: vCPU0’s trap returns before T1 releases the lock, and PCIe posted writes to one function are ordered. This is not exotic — it is **every cross-CPU register sequence RM orders with its own lock**.

⇒ **One MPSC register ring per device, its slot reserved by the same atomic that bumps `work_seq` (\$2.4).** The sequence is taken *inside* the trap, before it returns, so it already embeds the guest’s cross-vCPU happens-before; the ring is globally ordered by construction and no separate stamp is needed. The vCPU pays one atomic instead of two, and the per-vCPU claim bit disappears. A drainer that reaches a reserved-but-unpublished slot waits — bounded by one trap, microseconds, holding nothing.

∅ **This ring may not overflow, and that asymmetry is the whole point of \$2.1.** The doorbell table is *state* and a lost hint is recoverable by a scan; a register write is a *stream* and the stream **is** the truth — there is nothing to rescan and a dropped write is unrecoverable. So it must be sized so it cannot fill, and the bound comes from ogkm limiting in-flight entries. □ **That bound is asserted, not yet verified** — it belongs in \$10, because if it is wrong the failure is a guest that wedges with no diagnostic. Its saving grace is the privilege line: only guest root can reach it, and a guest root that overflows it harms only itself.

2.4 The wakeup protocol

One word, `worker_vcpu_poll`, and the futex `prepare_to_wait-then-schedule` idiom. `[owner]` supplied the field split; an adversarial review supplied the orderings and the layout.

```

worker_vcpu_poll : u64 = { work_seq : 56 (HIGH) , workers_polling : 8 (LOW) }

```

vCPU — publish the work, then `fetch_add(1 << 8) (AcqRel)`; if the pre-value's `workers_polling > 0`, `write(eventfd, 1)`.
Worker — `seen = work_seq (Acquire)` **before** scanning; scan every source; then CAS to register as polling **only if** `work_seq == seen (AcqRel)`; on failure, rescan.

□ Registering only when the sequence is unchanged is what deletes the “who clears the flag, and when” question that the flag-based version could not answer. Any work published after `seen` fails the CAS, so a worker cannot sleep on work it did not see.

Why 56/8 is right, with the number. ABA needs the counter to wrap *inside one worker's scan*, not globally — and every vCPU CASes the **same cacheline**, so aggregate bump rate is capped by ping-pong at roughly 10⁷/s regardless of vCPU count. 32 bits wraps in ~430 s (a descheduled worker could straddle it); 40 bits in ~30 h; **56 bits in ~228 years**. No spin-loop fallback is needed.

⊗ **work_seq must be the HIGH half.** With it low, the carry at 2⁵⁶ lands in `workers_polling` — a phantom poller, permanent, making every subsequent vCPU trap pay a `write()` syscall, silently. High, the carry falls off the top of the u64. The benign direction (a worker's `fetch_add(1)` carrying *into* `work_seq`) only causes a rescan.

□ **Cap the worker count at 255 structurally**, asserted at startup. `workers_polling` wrapping would read **0 while workers are polling** — precisely the lost wakeup the protocol exists to prevent. Saturating is worse than asserting: it makes the count wrong in a way that still reads plausible.

⊗ **Orderings are load-bearing and x86 TSO hides their absence.** With Relaxed on the vCPU's bump, the ring/table store may become visible *after* it on ARM: the worker scans empty, registers, sleeps, and the vCPU already read `workers_polling == 0` so sends no wake. Passes every bench in this tree; fails on the first Grace host. Rule: **every write to a work source happens-before the sequence bump, and every seen load happens-before the scan**.

⊗ **The eventfd is a sum, not a queue of wakeups.** M writes landing before the first `read()` collapse to one wakeup under `EPOLLEXCLUSIVE`, leaving N−1 workers asleep with work queued. ⇒ `EFD_SEMAPHORE | EFD_NONBLOCK`, **one shared epoll instance**, `read()` consumes exactly one. □ And a hard invariant: **no handler may wait on another queued item**. If one ever does, that collapse stops being latency and becomes a permanent hang.

⊗ **A bit that says work exists is not a bit that says you may touch the hardware.** Per token, two bits, four states:

```
vCPU:    fetch_or(RUNG)
worker:  CAS RUNG → BUSY      (fail ⇒ not ours, move on)
         act
         CAS BUSY → IDLE      (fail ⇒ state is BUSY_RUNG ⇒ CAS BUSY_RUNG → BUSY, act again)
```

Exactly one worker per channel at a time, no re-ring lost, no lock. □ Per §2.1 this is a **security** boundary: without it an unprivileged guest process triggers the concurrent walk of root's channel deliberately.

⊗ **Never clear RUNG on “cannot act yet.”** A doorbell can be acted on before a register write that preceded it on the same vCPU (different planes, §2.2 vs §2.3). On hardware the pending bit persists until the channel is schedulable; ours must too. Clearing it on a refusal is a lost doorbell — the same class as this tree's `forwarded=0 refused=8` engine-object rows.

⊗ **The claim is acquired INSIDE the scan, after seen, and released before the scan returns.** Caching “this ring was empty” across a seen read loses an item **forever**. This must be enforced by the API shape, not by a comment, because it is exactly the shape someone optimises.

□ **Instrumentation note.** A worker parked in a blocking `read()` counts in `workers_polling` but is **not** in `epoll_wait`. A census built on the word and one built on `epoll` state disagree only under load — the one condition in which anyone consults them.

2.5 The doorbell page is READ-ONLY, not emulated

□□□ The doorbell page is mapped into the guest as a **KVM read-only memslot** over the real host doorbell page. Consequences, all of them deletions:

- Every **read** is hardware, with **no exit and no code** — including the microsecond counter. ⊗ **There is no PTIMER implementation in v3.** The register emulation, the refusal of guest writes to it, the counter-page install — all gone.
- Every **write** traps, and runs §2.2.
- When we need to ring a doorbell (passthrough inline, or translated from a worker) we write the host page **directly**, bypassing the read-only mapping.

□ Emulated channels need no doorbell written at all — there is no hardware counterpart.

□ **[CORRECTED at w819] Passthrough and managed are different costs, and I had them merged.** A passthrough doorbell is a table lookup and one dword, synchronously, on the vCPU — **microseconds and a return**.

Only translated/emulated doorbells set a bit and wake. Translated and emulated behave identically from the vCPU's side; only passthrough is inline.

⇒ **This excludes ioeventfd for passthrough:** an ioeventfd would buy a cheap exit and then pay for it with a thread context switch to deliver a dword the vCPU could already have written.

3. Memory

Guest vidmem is **one host allocation**, and its size is a **command-line parameter of the hypervisor invocation**, exactly like guest RAM. **[CORRECTED]** v2 sized it by a bisecting probe and then rebound the advertised framebuffer to whatever the probe won; that is a measurement standing in for a decision. An operator asking for 12 GiB either gets it or gets a refusal at startup.

Guest FB offset **is** the offset into that allocation. Guest systemem is the hypervisor's memfd reached through a static layout.

4. Addresses

4.1 Mirror, never adopt — and the review's reasons are stronger than mine

[CORRECTED] I argued adopt was merely unsound because guest PTEs hold guest-physical addresses. The review gives three independent defeaters, any one of which is fatal:

1. You **race the host kernel module's own tables and invalidates**.
2. The host driver **exposes no userspace interface** to participate in its locking, so adopt needs a **host kernel module** — a deployment bottleneck for the whole product.
3. You need a translation **anyway**, because guest physical $\neq$ host physical. The only escape is a contiguous host range for guest physical, which then fails on **fragmentation** even when there is free memory.

⇒ Mirror is also **completely rootless**, which is the security posture the product needs.

4.2 The diff

The PTX walk kernel produces a **per-VA-space delta**: a list of VA spaces that changed, and the entries to apply to each. **∅ The previous state lives in vidmem, held by the PTX itself** — a snapshot of the full PD/PT tree per VA space, keyed by root address, with allocations freed when no longer referenced. **There is no host-side table of old pages.**

The delta is executed by the VA manager thread as mmap/munmap.

□ **Batching, and it is the answer to the syscall-cost question.** [owner] set the **TLB defer flag** — which ogkm demonstrably has — on every mapping call but the last. One invalidate for the whole batch instead of one per page.

4.3 A VA space is the OBJECT. The PDB is an attribute of it, per GPU.

∅∅∅ **[CORRECTED TWICE — and ogkm refutes the correction I accepted.]** I first wrote that a root is *only* a walk seed and never an identity. The review corrected me to (PDB base, aperture), and I agreed. **ogkm says neither of us was right, and the original Option<Pdb> design was.** Four independent reasons, all cited:

1. **RM's identity is the object plus vasUniqueId**, a monotonic atomic assigned at creation (virt_mem_mgr.c:143). Every runtime lookup is by client handle, never by PDB.
2. **One VA space has N PDBs on N GPUs** — the PDB lives in per-GPU state (gvaspaceGetPageDirBase(pGVAS, pGpu), gpu_vaspace.c:1874).
3. **A VA space can CHANGE its PDB, and can have NONE.** SET_PAGE_DIRECTORY is explicitly repeatable for resize/migrate (nvos.h:3084); gvaspaceResize migrates the root (gpu_vaspace.c:3235); and the same function reads if (NULL == gvaspaceGetPageDirBase(...)) goto doneGpu;. Hardware even encodes the absent case: NV_RAMIN_SC_PAGE_DIR_BASE_TARGET_INVALID.
4. **Nothing stops two VA spaces naming the same PDB** — there is no global PDB→VAS registry; RM's only check is local to the target VAS (gpu_vaspace.c:2855).

⇒ (PDB, aperture) **is a point-in-time HARDWARE FINGERPRINT, not a durable identity.** Use the VA-space object (its resource key) as identity; carry Option<(Pdb, Aperture)> as a mutable, per-GPU, possibly-absent attribute used to seed a walk and to match hardware state.

□ This retroactively vindicates the existing `Vas::pdb` doc — “None until a declaration arrives, and a space is nameable, routable and populatable before then” — which is exactly what `ogkm` describes. □ And it means my `Pdb(0)` fix treated a symptom: the durable fix is to stop using the fingerprint as the key at all.

4.4 Where the root actually comes from — it is ours to write

⊗ **Root discovery is not a problem, and the reason is stronger than “we can find it”.** `_gmmuWalkCBUpdatePdb` is a **no-op inside a guest or CPU RM** on GSP parts (`gmmu_walk.c:599`): *channel instance blocks are written by GSP-RM — which is us*. The guest RPCs the instance block’s physical address to GSP (`kernel_channel.c:2626`). We are the party that programs `NV_RAMIN_PAGE_DIR_BASE_{TARGET,VOL,LO,HI}`, and `TARGET` is the aperture, written as one unit with the address (`kern_gmmu_gp100.c:123`).

□□□ And `COPY_SERVER_RESERVED_PDES` is **the guest handing us real PDE physical addresses**. In the GSP split one logical VA space exists as *two* objects: client RM owns all page-table memory; **server RM — us — gets an exclusive 512 MiB window** (`0x1_0000_0000`, size `0x2000_0000`) plus the physical addresses, apertures and page shifts of the client’s PDE pages for that window, so the server can map into the same hardware tables without allocating any (`ctrl90f1.h:261`, `gpu_vaspace.c:4216`).

⊗ **DMA_SET_PAGE_DIRECTORY is UVM’s call, not a mapping call.** “DMA” is the *device’s DMA/VA-management sub-interface* of `NV01_DEVICE_0` — legacy taxonomy, and every sibling (`GET_PTE_INFO`, `INVALIDATE_TLB`, `SET_VA_SPACE_SIZE`) is VA-space-wide. What it does is relocate a VA space’s top-level page directory onto **client-supplied** memory; the caller is UVM at GPU-VA-space registration (`nv_gpu_ops.c:8986`). `UNSET` is its exact inverse.

⊗ **DUP_OBJECT aliases, it does not copy.** `vaspaceapiCopyConstruct` is literally `vaspaceIncRefCnt(pVAS); pVaspaceApi->pVASpace = pVAS; (vaspace_api.c:440)`. Two handles, one object — the normal UVM flow, not an anomaly.

5. Completions and interrupts

[CORRECTED] Most completion code deletes.

kind	what we do
Passthrough	nothing. We do not inspect the channel and do not care when it finishes.
Translated	nothing. The semaphore address was already forwarded; the GPU writes it.
Emulated	forge the completion at the end of our own call — there was no GPU work.

⇒ The completion-watch list, the 250 ms observer sweep and the CPU-executor completion writer all go.

What remains is the armed interrupt — and `ogkm` settles its shape.

□□□ **Registration is PER ENGINE, not per channel and not per semaphore.** `NV01_EVENT_NONSTALL_INTR` *requires* its notifier to be a **Subdevice**; the notify index maps to an engine (`NV2080_NOTIFIERS_CE0..n`, GR, HOST) and lands in `pGpu->engineNonstallIntrEventNotifications[rmEngineId]` (`event_notification.c:688`). The channel class `NVA06F` has **no completion event at all** — only error/RC notification. So the interrupt is a **broadcast wake to every waiter on that engine**, carrying no channel and no semaphore identity.

□□□ **The race is the WAITER’S burden, and it is handled three independent ways.** RM’s own semaphore-surface registration re-reads the value under a spinlock and returns `NV_ERR_ALREADY_SIGNALLED` — with the race written out in a comment (`sem_surf.c:1405`); callers loop on it (`nvidia-drm-fence.c:991`); and the OS-event wait path **never trusts the event**, looping on the notifier word in memory and using the event only to decide *when to re-check* (`nvidia-push.c:751`).

⇒ [owner’s hypothesis, confirmed] we only need to fire on **genuinely new completed work** — an advance of the engine. We do **not** have to fire retroactively for work already complete, because the guest re-checks after arming. □ But the converse is binding: once armed, a later release **must** produce an interrupt. There is no level-triggered delivery to fall back on.

⊗ **And the interrupt is requested in the PUSHBUFFER**, not by a control: `NV906F_NON_STALL_INTERRUPT` pushed after the release (`nvidia-push.c:967`), or `CE_LAUNCH_DMA ..._INTERRUPT_TYPE_NON_BLOCKING`, which RM sets **only when a completion callback was requested** (`channel_utils.c:657`; default `_NONE`). □ There is **no AWAKEN_ENABLE field** in the headers shipped here — an earlier note in this tree citing one should be re-read.

□ **The channels we most care about POLL anyway.** UVM never uses interrupts for channel completion — it spins (uvm_channel.c:2158). The CeUtils scrubber registers a non-stall event only for its *asynchronous callback* API; every blocking wait is while (READ_CHANNEL_PAYLOAD_SEMA(..) < target) { ... } (channel_utils.c:343), and when it holds the GPU lock it services the ISR **inline** rather than sleeping.

6. The scrub

□ **There IS a dedicated scrub channel, and a second one beside it.** OBJMEMSCRUB owns a CeUtils channel per heap, kind CE_SCRUBBER_CHANNEL or FAST_SCRUBBER_CHANNEL (SEC2's SWL_SCRUBBER_CHANNEL under Confidential Compute) — mem_scrub.c:171, ce_utils.c:246. A **separate** CeUtils instance serves MemoryManager's general internal copies (mem_mgr.c:4114), so scrubbing and other RM-internal CE work do not share a channel.

⇒ That makes the scrub a clean, named, **translated** channel. The guest's own RM scrubs before reuse and orders it correctly; our only job is to let the work reach the GPU instead of forging its completion. [measured w813] today the kernel tokens are forwarded=0 — 140 doorbells rung, none sent — which is the cross-client leak.

7. Portability — derive, do not hardcode

[owner] in priority order:

1. Derive from **host userspace calls** wherever possible.
2. **Computed** values are allowed.
3. Generate Rust from the **C headers** of ogkm and friends — dies, families, driver versions.
4. Read **nvkvm-pv** for how it already solved this.

⊗ **Do not parse C with regex or sed.** Use a real parser/compiler front end. [owner] “*That will age better.*” □ This tree's own history agrees: a capture-derived table expired as a vendor regression, and 11 of 56 captured rows were empty and every empty one that was checked against hardware was **contradicted**.

The surviving hand-written surface should be a small set of per-architecture-series constants.

8. What is deleted

Isolates and the whole IPC plane · the sandbox plane · Proc · the address table · joins (all four generations) · VA→phys translation on the submission path · the operand gate · publication epochs, the dirty gate, sweep-skip · the CPU CE executor · the completion watch · PTIMER emulation · the framebuffer probe/rebind · **every on/off flag for things that no longer have two sides** [owner].

□ [w820] **Three more, and they were in the v3 proposal itself — not in the old tree:**

- **The doorbell hint queue.** Superseded by the 64 KiB bit table (§2.2). Deleting it deletes ring overflow, the multi-consumer problem for doorbells, and an unprivileged guest process's ability to exhaust anything.
- **“Inspect all doorbells” / any FULL_REFRESH-shaped fallback.** There is no refresh because there is no queue to fall back *from* — there is only the scan, which is the normal mode. [owner, 2026-09-20] “*guest userspace can exist without something like FULL_REFRESH.*”
- **Per-vCPU register rings and their owner-claim bit.** Refuted in §2.3: they preserve per-vCPU order where the device needs the guest's cross-vCPU order.

⊗ Deleting a mechanism that only ever existed in a design document costs nothing and is the cheapest deletion available. □ Both doorbell deletions were **licensed by a threat model**, not by a benchmark — which is the only reason they were found before the code existed.

⊗ **[CORRECTED] What to keep is narrower than I said.** I proposed keeping “the instruments”. The review is right that some instrument *prose* is actively misleading — three stale “The default” comments in one day, one of which made a full survey report that the single store does not run. ⇒ Keep the **measurements** and the **gates**; audit the **prose**, and delete what can no longer be checked. A comment asserting a default will rot; a comment asserting a dated measurement will not.

9. Rewrite, not refactor

[owner asked for an honest view] **Rewrite, in a fresh crate set, with the old tree as a read-only reference.**

The deletions above are not edits inside the existing files — they are the removal of the organising ideas those files are built around (per-proc isolation, an address table, a publication pass). Deleting them incrementally means running two architectures in one file set for the duration, which is how the stale defaults and contradictory tests got there in the first place.

□ The one non-negotiable: **the rewrite must be graded continuously against the existing suite** — the 30-arm guest suite, the bare-metal baseline, and the hardware gate (forwarded= per token). A rewrite that cannot be graded is a rewrite that cannot be landed.

Ø [measured 2026-09-20] crates/*/src is **144,175 lines of code** and 125,503 lines of comment. A 50k target is a ~3x reduction of code, and the four crates carrying the deletions are kayfabe-isolate-host (46,895), kayfabe-abi (43,588), kayfabe-qemu-raw (37,057) and kayfabe-device (27,496). kayfabe-abi should shrink by being **generated** rather than deleted.

10. Still open

[w820] Resolved since w819, recorded here so the list does not read as unchanged: the doorbell/register queue split and its overflow semantics (§2.1-2.3); the worker wakeup protocol, its bit widths, its layout and its memory orderings (§2.4); the doorbell double-take, now a named security boundary rather than a race (§2.4).

Still genuinely open:

1. **The interrupt race** (§5) — per-channel or per-semaphore, and whose burden.
2. **The synchronous-verb list** — which RM controls does the guest read the reply to immediately? Until that list exists, “no RM verb on a vCPU” cannot be judged. □ **Owner is waiting on this one**; it blocks judging the trap path end to end.
3. **Fault delivery** — needed by managed memory *and* by letting the GPU fault. [owner] if the fault structures are shared with userspace we can map them through for passthrough; a translated fault needs its address translated on the way back.
4. **The crate model** — to be redrawn against §8.
5. □ **[NEW, w820] The privileged ring’s non-overflow bound is ASSERTED, not verified.** §2.3 needs it sized so it cannot fill, on the argument that ogkm limits in-flight entries. That bound must be **read out of ogkm and turned into a startup assertion**, because if it is wrong the failure is a guest that wedges with no diagnostic.
6. □ **[NEW, w820] The scan-cost figure in §2.2 is arithmetic, not a measurement.** Sub-µs warm is the claim the ring-deletion rests on; it needs a microbenchmark on the bench box before it is cited as fact.
7. □ **[NEW, w820] Enumerate every BAR0 range RM maps to a NON-kernel client.** §2.1 treats the usermode window as the unprivileged surface, which is true of ogkm today — but that is **RM policy, not an invariant**. The right form is a generated assertion over the objects that hand out ADDR_REGMEM, not a claim that there is only one. [owner] *“same for any other page if they exist.”*

Part 2 — The surface, at constant level

STATUS: LIVE, 2026-09-20 (w821). Companion to `THE_ARCHITECTURE_v3.md`. Supersedes the *data* in `THE_SURFACE_v2.md`; that document’s `KEEP/DELETE plans` still stand and are kept.

◊ **Written for a reader who does not know NVIDIA’s constants.** Every entry gives the constant name, its numeric value, one plain sentence saying what the thing *is*, and what we do with it. □ Every name and number is cited to `research_clones/ogkm` or to this tree. Anything a survey could not establish is marked **[UNVERIFIED]** rather than guessed.

0. The four planes, and why they are different shapes

A guest driver reaches a real GPU through exactly four kinds of traffic. *kayfabe* intercepts all four, and each has a different shape because each has a different **trust** and **ordering** requirement — not because of taste.

plane	transport	who can drive it	our posture
GSP RPC	a command/status ring in guest memory, kicked by a BAR0 register write	guest root (RM)	we are the GSP: we answer, forge or forward
RM control	carried <i>inside</i> GSP RPC GSP_RM_CONTROL, and via <code>ioctl</code> on the host side	guest root (RM)	serve, forward, refuse — by command number
BAR0 MMIO	reads untrapped from a DRAM shadow; writes trap to the vCPU	guest root , except the usermode window	allowlist (§41), shadow write-through
channel / pushbuffer	a GPFIFO ring in guest memory, kicked by a doorbell write	□ unprivileged guest userspace	passthrough / translated / emulated

□□□ **The last row is the one that restructured the design.** The doorbell is the only surface an unprivileged guest process touches directly, which makes it adversarial to the guest’s own root — see `THE_ARCHITECTURE_v3.md` §2.1 and `THE_CONSTRAINTS.md` §47.

⇒ Read the four sections below in that order: they go from most-privileged transport to least-privileged, and the machinery gets **weaker and more paranoid** as it goes down.

1. GSP RPC — the plane we answer as the firmware

1.1 What this plane is, for a reader who does not know NVIDIA’s constants

Modern NVIDIA GPUs run a firmware processor called the **GSP** (GPU System Processor). On such a part the host driver does not talk to the hardware directly for most things — it sends **messages** to the GSP, and the GSP’s own copy of the resource manager (**GSP-RM**) does the privileged work. *kayfabe*’s central trick is that **we are the GSP**: the guest’s stock, unpatched driver sends us these messages believing we are firmware.

The transport is a pair of rings in **guest system**, not a register interface:

- a **command queue** the guest writes and we read,
- a **status queue** we write and the guest reads,
- each ring made of fixed-size **elements**, each element carrying a checksum and a sequence number, each message an `rpc_message_header_v03_00` envelope plus a payload.

The guest tells us where those rings are once, at boot, through `MESSAGE_QUEUE_INIT_ARGUMENTS` (*kayfabe-gsp/src/boot.rs*:100, read at :1185). After that the **only** hardware event in the whole plane is a single register write that says “*the command queue moved*” — see §3.

1.2 The structures on the wire

structure	what it is	kayfabe [CODE]	ogkm [ogkm]	axis
msgqTxHeader	the ring's producer header: version, element size/count, write_ptr , flags, and the offsets of the rx header and the element array	TxHeader, kayfabe-gsp/src/ring.rs:131-148 (8×u32, 32 bytes, write_ptr at +16)	msgq_priv.h:37	stable
per-element header	prefixes every element: a checksum and a sequence number, plus transport framing	ElementLayout, kayfabe-gsp/src/element.rs:112-233	message_queue_priv.h:43 (580) / :52 (610)	Ø Dg — BREAKS
rpc_message_header_v03_00	the message envelope inside an element: header_version, signature, length, function, rpc_result, rpc_result_private, sequence — 32 bytes	RpcEnvelope, kayfabe-abi/src/view.rs:558-572	—	Dg (additive)
queue pair geometry	binds the guest's command queue and our status queue, including the swapped rx convention where each side publishes its consumption into the <i>other</i> queue's header	MsgqGeometry, kayfabe-gsp/src/ring.rs:407-570	MSGQ_FLAGS_SWAP_RX	stable

□□□ **The element header is the single best worked example of the Dg axis in this tree**, and it is why \$0.2 says descriptors rather than branches:

	580.159.04	610.43.02
header size fields	48 bytes authTagBuffer[16], aadBuffer[16], checksum@32, seqNum@36, elemCount@40	16 bytes mctpHeader, nvdmHeader, checksum@8, seqNum@12

⇒ Size *and* every field offset changed between two versions we must both support, and the elemCount field **disappeared**. kayfabe absorbs this as data — ElementLayout carries elem_count_off: Option<usize> and a TransportHdr that is None at 580 and Mctp{..} at 610 — not as a version branch. **[PROPOSE]** every other axis-carrying fact should take this shape.

1.3 The functions

Dispatch: FunctionCodes::classify(), kayfabe-gsp/src/rpc.rs:211-232. The number↔name mapping is **generated**, not transcribed: kayfabe-abi/src/generated/rpc.rs.

□ [ogkm] The enum (src/nvidia/inc/kernel/vgpu/rpc_global_enums.h) is **byte-identical between 580.159.04 and 610.43.02** for every id we use. 610 only *appends*. ⇒ On this plane the *numbers* are Dg-stable and only the *framing* moves.

#	NV_VGPU_MSG_FUNCTION_	what it is, in plain language	we do	[CODE]
1	SET_GUEST_SYSTEM_INFO	first message after boot: the guest states its driver version, page size and OS, and expects the agreed protocol version back	SERVE	kayfabe-device/src/guestsysinfo.rs:112
64	SET_GUEST_SYSTEM_INFO_EXT	follow-on carrying extra system strings; the real driver returns its status, so it must succeed	SERVE (NV_OK, empty body)	guestsysinfo.rs:122
65	GET_GSP_STATIC_INFO	the guest asks for the big fixed-fact table about this GPU — memory layout, engine list, BAR sizes — before anything else can initialise	SERVE from our device-data model	kayfabe-device/src/staticinfo.rs:182
70	UPDATE_BAR_PDE	the guest hands us the physical root of its BAR2 page directory. □ Only the firmware’s directory is hardware-rooted, so this is the only fact we get about that tree	SERVE (latch it)	kayfabe-device/src/bar2.rs:299
72	GSP_SET_SYSTEM_INFO	a system-description blob (PCI topology and friends), sent early, fire-and-forget	IGNORE — ∅ <i>no reply at all</i> ; echoing would desync the guest’s sequence counter	kayfabe-gsp/src/rpc.rs:318
73	SET_REGISTRY	the guest uploads its entire registry / module-parameter table	IGNORE (no reply)	same
10	FREE	destroy one object handle, or a whole client namespace — RM’s teardown stream	SERVE from the local object graph	kayfabe-rmrpc/src/lib.rs:1095, :1955
21	DUP_OBJECT	alias one client’s object into another client’s handle namespace. ∅ It aliases, refcount++ ; it does not copy — this is the normal UVM flow	SERVE (a graph edge)	lib.rs:1097, :1918
103	GSP_RM_ALLOC	the generic “ <i>create an RM object of this class</i> ” envelope. Nearly every object the guest makes — clients, devices, VA spaces, channels, memory, contexts — rides inside	MIXED, default REFUSE	lib.rs:1229-1276

#	NV_VGPU_MSG_FUNCTION_	what it is, in plain language	we do	[CODE]
76	GSP_RM_CONTROL	the generic “ <i>run this RM control command</i> ” envelope. Almost every specific query or operation rides inside as a nested NVxxxx_CTRL_CMD_* id + params — see §2	MIXED, default REFUSE	lib.rs:1521-1572
71	CONTINUATION_RECORD	Ø not a function: an overflow <i>fragment</i> of a message too large for one element. Only GSP_RM_CONTROL fragments this way	folded into reassembly; standalone is refused	kayfabe-rmrcp/src/reasm.rs:250
47	UNLOADING_GUEST_DRIVER	the guest is about to rmmmod. □ Synchronous — the guest blocks until we reply	SERVE (ack-only) , reply mandatory	kayfabe-device/src/inert.rs:119
202	ECC_NOTIFIER_WRITE_ACK	the guest’s own acknowledgement that it finished writing the ECC notifier buffer	IGNORE	rpc.rs:320
228	INIT_GSP_TRACE_CRASH_BUFFER	the guest hands us a {physical address, size} for a buffer the firmware is meant to fill with crash-trace records	SERVE (ack, no-op) — we write no trace records, so it stays zeroed	inert.rs:119
0x1001	GSP_INIT_DONE (<i>event</i>)	we tell the guest “ <i>boot finished</i> ”; its boot poll waits on this	outbound only	kayfabe-gsp boot FSM
0x1003	POST_EVENT (<i>event</i>)	the generic asynchronous-completion carrier firmware uses to notify the guest	outbound only	—
0x1004	RC_TRIGGERED (<i>event</i>)	we tell the guest a channel or engine was torn down by robust-channel fault recovery	outbound only	kayfabe-gsp/src/fault.rs
—	everything else (~210 ids)	any other function or event id	Ø REFUSE by name — NV_ERR_NOT_SUPPORTED (0x56), zeroed body	lib.rs:1146

□□□ **The safe default is a named refusal, never an echo.** A chain of CommandPolicy links is tried in order (PolicyChain::respond, kayfabe-gsp/src/boot.rs:596); whatever no link answers falls through to GspFsm::answer’s refusal (boot.rs:1719-1763). Ø An Echo0k policy exists (boot.rs:541) but is a **differential-test fixture reproducing the old C’s behaviour** and is **not installed in the production chain**. This matters because 0x56 is a status the guest driver forgives — so a generic-ack fallback would let a wrong configuration run on, undetected, which is precisely the failure this project has measured repeatedly.

1.4 □ The reply is always built from local state

∅ **No RPC is a verbatim forward.** Applying an `RmEvent` to the local object graph (`Gpu::apply`, `kayfabe-core/src/gpu.rs:5582`) can *latch* a pending host-side spawn, resolved immediately after by `materialize_pending` (`:5591`). Real host ioctls happen **asynchronously as a side effect**, decoupled from the synchronous RPC reply.

□ **That decoupling is load-bearing and was paid for:** it previously ran inline under the rank-0 lock and **crashed QEMU** (`gpu.rs:3462-3471`).

□ **[UNVERIFIED]** whether any individual modelled `GSP_RM_CONTROL` / `GSP_RM_ALLOC` arm blocks on a host call before constructing its reply. The top-level dispatch does not; the ~15 per-control handler files were not exhaustively checked. ⇒ This is the same question as the synchronous-verb list in §2.4, and it is the one the owner is waiting on.

3. BAR0 MMIO — reads from DRAM, writes to the vCPU

3.1 The asymmetry, and why it exists

BAR0 is the GPU’s register aperture: ~16 MiB of address space where each 4-byte slot is a control register. A normal emulated device traps **both** reads and writes, and pays a VM exit for each. `kayfabe` does not:

- ∅ **Reads are not trapped.** They are served out of **ordinary DRAM** the guest reads directly, with **no exit and no code running**.
- □ **Writes trap** into `nvkvm_trap_bar0_write` on the vCPU thread.

The reason is the trap contract: a vCPU inside an MMIO exit is **not preemptible**, so every trapped read is a stall charged to a core of the customer’s VM. A register the guest *polls* — and driver init polls several thousand times — would be thousands of stalls for a value that did not change. ⇒ Put the value in memory and let the guest read it at DRAM speed.

[MEASURED] **3 572 of 4 096 BAR0 pages are backable in 12 contiguous runs** (`kayfabe-device/src/plane.rs:2560`), against a read census of **241 722** BAR0 reads.

3.2 The shadow — two structures, and they are not the same thing

□ This is the part most often got wrong: “*the shadow*” names **two** objects.

	RegPlane	ShadowSink / ShadowSegment
what	the model : the source of truth that answers a <i>trapped</i> access	the DRAM the guest actually reads
[CODE]	<code>kayfabe-device/src/plane.rs:1223</code>	<code>kayfabe-qemu-raw/src/shim_unsafe.rs:1656, :1626</code>
shape	<code>ChipProfile</code> + <code>GspModel</code> + ROM image + two <code>RankedMutex</code> state cells + a dead-page bitmap	a <code>Vec</code> of { <code>off</code> , <code>len</code> , <code>base</code> } contiguous byte runs, linearly scanned
guest touches it?	∅ never	□ directly, with no exit

∅ **It is not a flat offset-indexed array and not a page hash map.** It is a small vector of merged contiguous runs — which is why a write to an offset no segment covers is **silently dropped** rather than faulting (`shim_unsafe.rs:1666-1677`).

How a page becomes readable-from-DRAM. `bar0_backable_runs()` (`plane.rs:2579`) sweeps the whole aperture in 4 KiB units and admits a page only if **every dword in it** is answerable without running code: a provably dead dword, a boot-register constant, a ROM byte, a CPU interrupt-tree leaf, the BAR0-window latch, an MMU-invalidate register, or something the GSP model can project. Adjacent admitted pages merge into runs.

□ **The content is computed, never loaded from a capture.** `bar0_shadow_fill` (`plane.rs:2743`) fills each run from the *same* per-arm sources the trapped read path uses, and only then does the shim attach that memory (`:1713`). ⇒ There is no capture table to expire as a vendor regression, which is the failure mode §0.3 forbids.

Write-through is a single funnel. `RegPlane::shadow_write` (`plane.rs:3144`) is the only producer, called from the BAR0-window latch, the MMU-invalidate latches and trigger, the interrupt-tree republish, and the GSP register-group republish (`:3933-4013`).

□ **[UNVERIFIED — a discrepancy worth someone’s attention, not yet a confirmed defect.]** Under `feature = "host-isolates"` a second mechanism, `Regs::back_bar0_dead_runs` (`kayfabe-qemu-raw/src/shim.rs:17527`), independently installs a read-only KVM memslot over `bar0_backable_runs()` backed by a `memfd` its own comments describe as “*never*

written... zero resident pages” — **without calling bar0_shadow_fill first**. Its prose still carries the pre-w563 *dead-pages-only* meaning while calling the post-w563 *widened* range, which also includes non-zero boot registers and ROM. The test at `crates/kayfabe-device/tests/bar0_backable_pages.rs:143` explicitly records that w563 changed the invariant from “*the run answers zero*” to “*the run answers what bar0_shadow_fill computes.*” ⇒ Either the two paths are gated to disjoint builds, or one of them serves zeros where the guest expects NV_PMC_BOOT_0. **The call ordering crosses the FFI boundary into the C shim, which is not in this repository, so it could not be settled from here.**

3.3 The trapped-write register table

RegPlane::write, plane.rs:6473. First match wins; the order is itself load-bearing (the doorbell is classified **before the plane lock is taken**, because its port may block).

Offsets are GA10x/GA106. **Axis** column: what moves the offset or the meaning.

offset	register (ogkm swref)	what it is, in plain language	what we do	axis
0x001700	NV_PBUS_BAR0_WINDOW	a relocatable 1 MiB window latch: bits pick <i>which</i> megabyte of framebuffer the PRAMIN aperture below currently points at	store the raw word verbatim (a read-modify-write must see back exactly what it wrote); write through to shadow	A
0xBB0090	NV_VIRTUAL_FUNCTION_DOORBELL	the bell : a work-submission token saying a command ring has new entries	§2 of Part 1 — the security-critical path. Classified with no lock held	Ø die
0xB830A0	..._PRIV_MMU_INVALIDATE_P10	low/aperture half of <i>which page directory to invalidate</i>	latch only, not acted on	A
0xB830A4	..._PRIV_MMU_INVALIDATE_U10	high half of that address	latch only	A
0xB830B0	..._PRIV_MMU_INVALIDATE_TRIGGER	the TLB-invalidate fire register: the guest writes scope + TRIGGER=1, then spin-polls this same register until TRIGGER reads back 0	decode scope, publish address-space changes here (the guest is quiesced — this is the exact GPU boundary), then clear TRIGGER unconditionally . ☐ A stuck TRIGGER hangs or resets the guest	A
0xB81000+i*4	..._PRIV_CPU_INTR_LEAF(i)	pending-interrupt bitmap, 32 vectors per leaf, write-1-to-clear	update tree, republish the whole group to the shadow, maybe raise an interrupt. No plane lock	A
0xB81200/1400	..._LEAF_EN_SET/_CLEAR(i)	enable mask for leaf i, set and clear ports	same arm	A
0xB81600/1608/1610	..._TOP(i), _TOP_EN_SET/_CLEAR	top-level pending summary and its enables	same arm	A
0xB81640	..._LEAF_TRIGGER	software interrupt injection (write-only)	same arm. ☐ tested first , to avoid aliasing with the TOP_EN_CLEAR array bound	A

offset	register (ogkm swref)	what it is, in plain language	what we do	axis
0x110C00+i*8	NV_PGSP_QUEUE_HEAD(i)	the GSP command doorbell : “a new command is in the ring”. This is the whole hardware surface of the RPC plane in §1	post and return — increment a counter, no lock, no inline drain. [MEASURED] this fixed a 1.79 s vCPU stall (plane.rs:5196). Drained later by a worker, one command at a time	A, see below
0xBB0080/0084	NV_VIRTUAL_FUNCTION_TIME	the GPU’s free-running nanosecond timer, read by the driver for timeouts	refused by name and counted — accepting a write would put guest and host GPU on different timebases	die
0x118234	NV_PGC6_AON_SECURE_SCRATCH	firmware (boot-progress scratch byte the driver polls for “GFW boot complete”	drive the boot FSM; republish group	A
0x118128	..._GROUP_05_PRIV_LEVEL_MASK	privilege mask that must be fully lowered before the guest trusts the progress byte	same	A
0x110100	NV_PFALCON_FALCON_CPUCTL (GSP)	the bit that starts the GSP microcontroller running	boot FSM; is_startcpu() classifies	A
0x1100F4	..._HWCFG2 (GSP)	reports/gates whether the falcon’s RISC-V core is enabled	boot FSM	A
0x110118	..._DMATRFCMD (GSP)	the ucode-load DMA command/status register	boot FSM	A
0x110040/0044	..._MAILBOX0/1 (GSP)	the two halves of the LibOS boot-args physical address	boot FSM	A
0x110008/0018/001C	..._IRQSTAT/IRQMASK/IRQDEST (GSP)	falcon interrupt status, mask, routing	boot FSM	A
0x110004	..._IRQSCLR (GSP)	write-1-to-clear falcon interrupt status. the guest’s ISR clears here before draining our status queue	boot FSM; is_swgen0_clear() classifies	A
0x111388	NV_PRISCV_RISCV_CPUCTL	RISC-V core control; the ACTIVE/HALTED bits the guest polls for liveness	boot FSM	A
0x111528/152C	NV_PRISCV_RISCV_IRQMASK/IRQDEST	RISC-V interrupt mask and destination, ANDed for stall attribution	boot FSM	A
0x840100	NV_PFALCON_FALCON_CPUCTL (SEC2)	start register of the security co-processor that loads protected firmware	boot FSM	A

offset	register (ogkm swref)	what it is, in plain language	what we do	axis
0x840040	..._MAILBOX0 (SEC2)	the Booter's argument — distinguishes a Load from an Unload of the protected region	boot FSM; is_booter_unload()	A
0x840118	..._DMATRFCMD (SEC2)	SEC2's ucode-load DMA command/status	boot FSM	A
0x1FA824/828	NV_PFB_PRI_MMU_WPR2_ADDR_LO/	base of WPR2 , the write-protected region where GPU firmware lives. $\square$ HI $\neq 0$ is the guest's "is WPR2 already up" test	boot FSM	$\emptyset \emptyset$ die $\times$ Dg — see below

$\emptyset$ **Everything not in this table is silently dropped** — counted in `unclaimed_writes`, returning `WriteOutcome::nothing()` (`plane.rs:6733-6737`). That includes writes to boot-register constants and to the ROM window, which are read-only from the guest's side and have no write arm at all.

3.4 $\square\square\square$ WPR2 is the worked example of two axes crossing in one register

`gb20x.rs:196-227`: on Blackwell the register **moved** to `0x0088A824/828` (`NV_HUBMMU_PRI_MMU_WPR2_ADDR_LO/HI`) for 610-series drivers — **while still answering the legacy `0x1FA824/828` pair for 580-series drivers**. Both offsets decode to the same internal variant.

$\Rightarrow$ **One register, two axes at once**: the **die** moved it, and the **guest driver version** decides which address that guest will use. Neither axis alone explains the behaviour, and a design that modelled only "*per-die offsets*" would answer the wrong address for a 580 guest on a Blackwell part. $\square$ This is the concrete argument for \$0.2's claim that axes are not independent and a per-die table is not sufficient on its own.

$\square$ Related, and **[UNVERIFIED]**: `NV_PGSP_QUEUE_HEAD` was once believed Turing-only; `gh100.rs:51` records the correction that **GH100's driver does write it**. $\Rightarrow$ An "architecture X doesn't use this register" belief is exactly the kind of claim that expires.

3.5 Whole apertures, not registers

range	name	treatment
0x700000-0x7FFFFFFF	PRAMIN , 1 MiB	one aperture , classified before the register decode. Every dword resolves through the <code>0x1700</code> latch: <code>fb_addr = (base << 16) + window_off</code> , then reads/writes the guest framebuffer store, or refuses by name if untranslatable. $\square$ A bring-up aperture, not a running path
0x300000-0x3FFFFFFF	PROM/VBIOS , 1 MiB	one range on read only , served from a byte image. $\emptyset$ No write arm exists at all — writes fall to the dropped path
0xB80000-0xB8FFFF	usermode / VF window , 64 KiB	$\emptyset$ not a pass-through aperture — decoded register by register (doorbell, invalidate trio, interrupt tree, PTIMER). Any other offset inside it is dropped

range	name	treatment
MSI-X table / PBA	—	Ø not in this tree at all . Only a vector <i>count</i> lives here; the table and delivery are QEMU’s native model on the C side. [UNVERIFIED] whether it sits inside BAR0’s range
BAR1 / BAR2	framebuffer / instance windows	separate PCI BARs, Ø never trapped — ensuring that is a requirement (Part 1 §2)

4. Channels and pushbuffers — the plane unprivileged guest code drives

4.1 What a channel is, for a reader who does not know NVIDIA’s constants

A **channel** is one command stream to the GPU. It has three pieces of memory:

- a **pushbuffer** — a byte array of commands (“methods”), written by the guest;
- a **GPFIFO ring** — an array of 8-byte entries, each pointing at a run of pushbuffer;
- **USERD** — a small per-channel block holding the ring cursors GP_PUT (producer, written by the guest) and GP_GET (consumer, advanced by the hardware).

To submit work the guest appends a GPFIFO entry, bumps GP_PUT, and **rings the doorbell**. The hardware’s front-end then reads GP_PUT, fetches entries, and executes the methods.

□□□ GP_PUT lives in USERD, which is mapped to the owning process and nobody else. That is the whole reason a hostile doorbell is harmless on real hardware and dangerous for us: hardware re-reads memory only the owner can write, and we *parse the owner’s pushbuffer*. See Part 1 §2.1.

4.2 The three kinds — and an honest statement of what is built

kind	contract	what it means
Passthrough	RingAndReturn — “no inspection and no work”; may run on the vCPU	the guest’s ring is in real GPU memory; hardware fetches it directly. We never read a byte of it
Translated	ScheduleAndReturn; must not run on the vCPU	every entry’s operands rewritten into our own VA space, then submitted on our host channel. The GPU moves the bytes
Emulated	ScheduleAndReturn; must not run on the vCPU	we implement the function in the VMM

[CODE] GuestChannelKind, kayfabe-core/src/channel_kind.rs:96-158; contracts at :286-315.

ØØØ **The classifier is a two-way switch, not a three-way one, and Translated is never constructed.** ProcBoundary::channel_kind() (kayfabe-core/src/project.rs:308-316) is:

```
if self.anchor == SYSTEM_ANCHOR { Emulated } else { Passthrough }
```

⇒ A channel is **Emulated** iff its client namespace is the one reserved *system* component (the guest **kernel’s** own client — UVM, the CeUtils scrubber), and **Passthrough** otherwise. □ **No source location tests a class id to decide the kind.** The class id only matters later, for method decode and engine routing.

Ø GuestChannelKind::Translated is a fully-typed third variant with complete contracts and exhaustive-match coverage everywhere — and **zero construction sites outside tests** (channel_kind.rs:355 is a test fixture array). The design doc that introduced it (the_three_channel_kinds.md, STATUS: DESIGN, w803) says “not yet implemented”, and that is still true. ⇒ **Real kernel-CE work the design says should be Translated is today classified Emulated** and driven through the CPU/host-CE forwarding path in §4.5. □ This is the gap behind the scrub defect in Part 5.

class (name + hex)	what it is	kind today	axis
AMPERE_CHANNEL_GPFIFO_A 0xC56F — user CUDA context	a userspace channel a CUDA process owns	Passthrough	A
AMPERE_CHANNEL_GPFIFO_A 0xC56F — guest-kernel / UVM / CeUtils	a channel the guest kernel driver owns	Emulated (<i>design target: Translated</i>)	A
AMPERE_DMA_COPY_B 0xC7B5	the copy engine class object — does DMA copies and memsets	inherits its channel's kind	A
AMPERE_COMPUTE_B 0xC7C0	the compute class object — kernel launches	inherits; routed HostGr, real hardware runs it, no in-process executor	A
GP100_UVM_SW 0xC076	∅ not a real engine . A software placeholder so UVM's channel allocation succeeds, and a subchannel to hold fault-cancel methods	rides its (kernel) channel → Emulated	stable
VOLTA_CHANNEL_GPFIFO_A 0xC36F, TURING_..., HOPPER_..., 0xC86F, BLACKWELL_... 0xC96F	the same channel object, per architecture	kind is per-namespace, not per-class	∅ A
AMPERE_USERMODE_A 0xC361	∅ not a channel — the BAR window object that exposes the doorbell register itself (Part 1 §2.1)	n/a	A

4.3 The GPFIFO entry — 8 bytes

[CODE] GpfifoEntry, kayfabe-abi/src/submit.rs:1836; decode at :1852. [ogkm] NVC56F_GP_ENTRY0/1, clc56f.h:262-284.

field	bits	meaning
GP_ENTRY0_FETCH	0:0	unconditional / conditional fetch (not modelled)
GP_ENTRY0_GET	31:2	pushbuffer address, bits 31:2
GP_ENTRY1_GET_HI	7:0	address bits 39:32
GP_ENTRY1_LEVEL	9:9	main run, or a subroutine call
GP_ENTRY1_LENGTH	30:10	length in dwords
GP_ENTRY1_SYNC	31:31	PROCEED, OR WAIT — the front-end must drain before fetching

□ LENGTH == 0 is **not an empty run** — it is a *control* entry whose low byte is an opcode, so GET_HI must not be read there. We refuse it rather than decode it (submit.rs:1822).

□ **Axis note:** NVC36F (Volta) and NVC56F (Ampere) differ in exactly two bits across the whole entry — C56F **dropped** PRIV (8:8) and **gained** INVALID_SCOPE in MEM_OP_A. = This encoding is far more A-stable than the register offsets in §3. We model neither bit.

4.4 The method header — one dword

[CODE] method_header_decode, submit.rs:2196. [ogkm] NVC56F_DMA_*, clc56f.h:286-360 — bit-identical to NVC36F_DMA_*.

field	bits	meaning
SEC_OP	31:29	which of 8 encodings this dword is
METHOD_COUNT / IMMD_DATA	28:16	dwords that follow, or the immediate payload itself
METHOD_SUBCHANNEL	15:13	which of 8 bound engine objects this addresses
METHOD_ADDRESS	11:0	□ dword-indexed — multiply by 4 for a byte offset

SEC_OP values: GRP0_USE_TERT(0), INC_METHOD(1), GRP2_USE_TERT(2), NON_INC_METHOD(3), IMMD_DATA_METHOD(4), ONE_INC(5), RESERVED6(6), END_PB_SEGMENT(7).

□□□ **Sizing and decoding are deliberately different jobs.** We *size* (skip correctly past) all eight forms, so the stream never desynchronises; we *decode the payload of* only incrementing runs. ∅ RESERVED6 has no defined format, so it decodes to nothing rather than to a guess. **[PROPOSE]** this split — *always know the length, decode only what you model* — is the right shape for every parser in this system, and it is what lets an unmodelled method be harmless.

4.5 Every pushbuffer method we recognise

method (ogkm constant)	offset	engine	what it does	what we do
NVC56F_SET_OBJECT	0x000	host FIFO	binds a class to a subchannel — this is how the stream says “ <i>the next methods are for the copy engine</i> ”	bind/clear per-subchannel decode state (ga10x.rs:1762)
NVC56F_SEM_ADDR_LO/_HI, SEM_PAYLOAD_LO/_HI, SEM_EXECUTE	0x5C-0x6C	host FIFO	the channel’s own semaphore release : write a payload to a GPU address when this point in the stream retires. This is how the guest learns work finished	decoded only for OPERATION == RELEASE; observed on the completion queue (ga10x.rs:1160)
NVC56F_MEM_OP_A/B/C/D	0x28-0x34	host FIFO	the TLB invalidate transport, targeted at one page directory	decoded for MMU_TLB_INVALIDATE/_TARGETED with PDB_ONE; counted, and the membar honoured as a hard barrier. ∅ no real invalidate is issued (ga10x.rs:1558)
NVC7B5_OFFSET_IN_UPPER/_LOWER SET_OUT_UPPER/_LOWER	0x200-0x40C	CE	source and destination addresses of a copy	latched (ga10x.rs:1195)
NVC7B5_LINE_LENGTH_IN, LINE_COUNT	0x418, 0x41C	CE	how many bytes per line, and how many lines	latched
NVC7B5_SET_SRC/DST_PHYS_MODEL	0x600-0x674	CE	which physical aperture (coherent system / non-coherent system / vidmem) a physical operand names	latched; read only when that operand is physical
NVC7B5_SET_REMAP_CONST_A/_B, SET_REMAP_COMPONENTS	0x700-0x708	CE	the constant-fill / component-swizzle parameters — this is how a memset is expressed	latched, consumed by remap decode
NVC7B5_SET_SEMAPHORE_A/B, _PAYLOAD[_UPPER]	0x240-0x24C	CE	the copy engine’s own completion semaphore address and payload	latched, consumed inside LAUNCH_DMA

method (ogkm constant)	offset	engine	what it does	what we do
NVC7B5_LAUNCH_DMA	0x300	CE	□ starts the copy. Carries no operands — every operand was latched by the methods above; the method word itself is all flags	the whole execute decision (ga10x.rs:1453, kayfabe-fwd/src/lib.rs:8724). See below
NVC076_SET_OBJECT, NO_OPERATION	0x000, 0x100	UVM SW	routine bind / no-op on the software class	ordinary; explicitly excluded from the tripwire
NVC076_FAULT_CANCEL_A/B/C, 0x104-0x114 CLEAR_FAULTED_A/B		UVM SW	UVM cancelling or clearing a real MMU fault	◊ refuses the whole parse — UvmFaultMethodWithoutFaultDelivery (lib.rs:8919). We have no fault delivery, so seeing one means the assumption behind admitting this class broke. □ A tripwire, not an error falls through to opaque
NVC36F_NON_STALL_INTERRUPT	0x020	host FIFO	◊ not recognised anywhere. grep across all crates returns nothing	
everything else	—	—	WFI, YIELD, NOP, CRC check, sub-device masks, legacy forms, non-incrementing and immediate runs	sized, not decoded — counted as opaque, stream stays in sync

□ **NON_STALL_INTERRUPT being unrecognised is a real gap against Part 1 §5.** That section says the interrupt is *requested in the pushbuffer*, and this is the method that requests it. We currently do not see it. ◊ Today that is survivable only because the channels we care about **poll** — UVM spins, and the scrubber’s blocking waits loop on the semaphore word — but it is an assumption about the guest, not a property of our design.

□ **Why LAUNCH_DMA is not in the single-method decoder:** a real copy is **five separate method runs**, and LAUNCH_DMA carries none of the operands. Only the stateful walker that threads an accumulator across the whole run can produce a copy fact (ga10x.rs:1730). **[PROPOSE]** this is the general shape — *a method is not a unit of meaning; a run is* — and any redesign that decodes methods individually will reintroduce this bug.

4.6 What happens when each kind’s doorbell rings

Passthrough. The host channel was adopted at **channel allocation**, never lazily (plan_channel_birth, lib.rs:5576) — □ mandatory, because RM **zeroes a caller-supplied USERD at alloc time**, so adopting at first doorbell would wipe the cursor that just rang. The doorbell resolves the guest token to the already-adopted host token, runs the ring gate, and writes the real host doorbell. ◊ **No pushbuffer byte is ever read** — device.rs:4110 says so by name, and ring_content_is_forwardable returns false for this kind. Completions are hardware’s own writes to guest memory; we observe nothing and forge nothing.

Emulated. Born lazily at first doorbell if needed. Handed to a worker — must not run on the vCPU. If the bound engine routes to CpuCe, the ring is parsed: read the guest’s pushbuffer, decode runs into methods, then for each LAUNCH_DMA partition the copy into spans by address representability, classify each as a **data copy** (a VA operand) or a **page-table write** (a physical operand — latched and applied to the *owning* process’s VA space, which may not be the issuer). Host-CE spans go to the real copy engine and **the real engine writes the guest’s completion**. If the engine routes HostGr, the doorbell is forwarded but ◊ **the ring is never parsed** — “no executor IN THIS PROCESS runs GR work” (device.rs:8930).

◊◊◊ **The measured defect, in one line:** CPU-executed spans have **zero completion writers** — write_completion has no call sites (lib.rs:8862), so a completion on that arm is **silently dropped**. Combined with the scrub being

classified `Emulated` rather than `Translated`, this is the cross-client leak in Part 5.

Translated. $\otimes$ Design only; no doorbell reaches it. The intended path is in `the_three_channel_kinds.md` §4: read `USERD`, advance `GP_GET`, bounded-copy the pushbuffer, rewrite every operand aperture `PHYSICAL` $\rightarrow$ `VIRTUAL` into one static VA space, submit on our host channel, and translate the completion **in address only** — the real engine writes the value. An untranslatable operand is meant to be forwarded as a **deliberate fault at a named sentinel VA**, never emulated and never silently dropped.

Part 3 — The machine, as it is

STATUS: LIVE, 2026-09-20 (w816). Companion to `THE_ARCHITECTURE_v2.md` (the proposal) and `THE_SURFACE_v2.md` (the per-item inventory). **This** document is the one an engineer should attack: it describes the system as it actually runs, from a file:line survey, and marks every place I propose to change it.

0. How to read this

Three markers, used consistently:

marker	meaning
[MEASURED]	a number from a real boot, with the run that produced it
[CODE]	a fact read out of the tree at w749-fable-legb, with file:line
[PROPOSE]	my design opinion. Argue with these.
□ [UNVERIFIED]	the survey could not establish it. Not filled in with a guess.

∅ Two contradictions found while writing this are left **in**, unresolved, because only one side of each can be true and I could not determine which:

1. **The BQL.** `shim.rs` and `docs/design/blocking_and_completion_model.md:16` both state “*every guest MMIO write arrives with the QEMU BQL held.*” But `qemu/hw/misc/nvkm/nvkm.c:1118` calls `memory_region_enable_lockless_io` on **every** region, the realize-time self-check *refuses the device* if the flag is missing (`nvkm.c:1568`), and `nvkm.c:392` states as measured fact: “*these callbacks run CONCURRENTLY on every vCPU with NO BQL.*” □ If the lockless path is live, every “the whole VM is frozen during a trap” argument in this tree is stale, and the shared-lock findings in §4 get much sharper.
2. **Vas is keyed by (GpuId, ResourceKey)** [`CODE gpu.rs:1761`], but at least three doc comments still say (GpuId, Pdb) (`gpu.rs:1760, :250, :275`). □ I did not determine whether any reachability argument still depends on the stale key.

□ And a third, which I fixed while writing: `ScratchpadArm::Off` is documented “*The default*” while `scratchpad_from(None)` returns `Measure`. That stale line made one of my own surveys report that the single store is off by default — i.e. that the central memory design does not run. It is the **third** stale-default of the day, after `IsolatePlane::Stillborn` and `KAYFABE_VAS_OWNER=isolate`. ## 1. Vocabulary — what the nouns actually are

∅ Written for a reader who does not know NVIDIA’s constant names. Every term below appears in the rest of the document and in the code; this is the only place they are defined.

GPU, host, guest. The *host* is the machine with the real card. The *guest* is a VM running a **stock, unmodified** NVIDIA driver. The guest believes it owns a GPU. It does not — it owns our emulation of one, and real work is carried out on the host’s card by unprivileged host processes we control.

RM (Resource Manager). The core of NVIDIA’s driver. Everything the driver does is “allocate an object of class X under parent Y”, “call control command Z on object W”, “free it”. Objects form a tree rooted at a *client*. This object protocol is the control plane, and it is what we emulate.

Client / handle / object. A *client* is a driver-level session (roughly: one open of `/dev/nvidiactl`). A *handle* is a 32-bit number the guest picks to name an object **within** its client. Handles are per-client, reusable after free, and therefore not identities — two different objects can wear the same number at different times. This is why the code keeps “incarnations” rather than trusting handle values.

GSP (GPU System Processor). On modern cards, a RISC-V core **on the GPU** runs a firmware that does most of RM’s privileged work. The host driver stops touching hardware directly and instead sends the GSP messages over a queue in memory. □ **We impersonate the GSP.** That is the central trick: we do not emulate a GPU register-by-register, we answer the driver’s *messages* the way its firmware would.

RPC. One of those messages. Each has a numeric function id (e.g. `0x67` = “allocate an object”). The guest writes a message into a ring buffer in its own memory and pokes a register; we read it, act, and write a reply back into the ring.

Control command. A sub-kind of RPC: “run command `0x20800102` against object H, here are the parameters.” These are the bulk of the driver’s vocabulary — hundreds of them, each with its own parameter struct. Most are *queries* (“how much framebuffer is there?”, “what are this copy engine’s capabilities?”).

BAR0 / BAR1 / BAR2. PCI memory windows the card exposes to the CPU. - **BAR0** is the register aperture — 16 MiB of control registers. A guest read or write here traps to us. - **BAR1** is a window onto video memory, so the CPU can read/write VRAM. - **BAR2** is a second window, used for *instance memory* (the structures describing channels).

Framebuffer / VRAM / vidmem. The card’s own memory. A *framebuffer offset* is a byte offset into it. □ In our design guest vidmem is **one** big host allocation, and a guest framebuffer offset is simply an offset into that allocation — the identity is the entire memory model.

Systemem. Ordinary system RAM the GPU can also read, via an IOMMU-ish mapping. For us this is the guest’s RAM, which the hypervisor hands us as a shareable file descriptor (`memfd`).

GPU virtual address (VA) and the page tables. The GPU has its own MMU. Work submitted to it uses *virtual* addresses, translated by page tables the driver builds in memory. A **PDB** (page directory base) is the physical address of the root of one such tree — i.e. “which address space”. The GPU’s TLB caches translations, so after changing tables the driver issues a **TLB invalidate**.

VA space. One page-table tree = one address space. A guest process typically has one; the kernel driver has others. Two channels sharing a VA space see identical translations.

Channel / GPFIFO. A *channel* is a submission queue to an engine. The driver writes command packets into a **pushbuffer** in memory, then appends an entry to a ring called the **GPFIFO** pointing at that pushbuffer. Hardware consumes the ring.

USERD. A small per-channel structure holding the ring cursors: `GP_PUT` (how far the driver has filled) and `GP_GET` (how far hardware has consumed). The driver writes `GP_PUT`; hardware writes `GP_GET`.

Doorbell. After updating `GP_PUT`, the driver writes one 32-bit word to a register to say “there is new work on channel N”. That write is a *doorbell*. It is the hottest trap in the system — and on a real card it is a posted write that costs the driver almost nothing.

Engines: CE and GR. - **CE (Copy Engine)** — a DMA engine. It moves bytes between addresses. `LAUNCH_DMA` is its “go” command; operands (source, destination, length) are loaded into it by preceding commands. - **GR (Graphics/Compute)** — runs shaders/kernels. CUDA work lands here.

Semaphore release. How software learns an engine finished. The driver puts a “when done, write value V to address A” command in the pushbuffer, then **spins** reading address A until it sees V. ∅ This matters enormously for us: the wait is the *guest’s own* spin on *its own* memory, so if the value appears the guest proceeds, and if it never appears the guest hangs. We must never forge V for work that genuinely had to happen.

Scrub. Before RM hands a page of video memory to a different client, it clears it — with a copy-engine fill, on a kernel-owned channel. This is what stops one process reading another’s old data. □ The guest does this itself and orders it correctly; our only job is not to lie about whether it happened.

WPR2 (Write-Protected Region 2). A region of VRAM the GSP firmware is loaded into, fenced off by hardware. The driver checks registers to see whether it is already up. It only resets on a full device reset, which is why each clean run needs a fresh VM boot.

CeUtils. RM’s internal helper that owns the kernel-side copy-engine channels — the ones used for scrubbing and other housekeeping. Its channels belong to no guest process.

UVM (Unified Virtual Memory). A second guest driver module (`nvidia-uvm`) providing managed memory. It has its own ioctl surface and its own channels, and CUDA uses it heavily.

Xid. An NVIDIA error code reported by the driver when hardware faults. `xid 31` is a GPU memory-access fault — the engine touched an address the page tables did not map. ∅ For us an Xid on the host is a signal we got a mapping wrong.

2. Processes and threads

2.1 The processes

A full Mode-2 boot is **one QEMU process** with our Rust archive statically linked in as a QOM device, plus **N+1 sandboxed child processes**.

process	count / lifetime	opens <code>/dev/nvidia*?</code>	what it is for
VMM (QEMU)	1 per VM	No [CODE]	owns the guest, KVM fd, all guest RAM, the MMIO traps, the pure core, the doorbell table, the BAR mirrors, and the isolate factory

process	count / lifetime	opens /dev/nvidia*?	what it is for
Scratchpad isolate	1 per (VM, GPU), spawned at PCI realize	Yes	holds the ONE reserved video-memory object, the walk kernel and (when armed) a CUDA context that guest process's RM client, its channels, its VA spaces
Per-proc isolate	1 per (guest process, GPU), lazy	Yes	

[CODE] Every `/dev/nvidiactl` open in the workspace is in `kayfabe-isolate-host` (the child side) or the standalone ladder tool. **The VMM issues no RM ioctls.** It holds descriptors the children hand up — sealed memfds and armed device nodes — which it only `mmap`s, never `ioctl`s (`export.rs:31-42`).

□ **The “birth client B” is not a process.** It is an RM client handle plus two character-device descriptors, minted **inside a per-proc isolate** (because RM stamps `pClient->ProcID` from the creating task) and passed to the scratchpad by `SCM_RIGHTS` [CODE `child.rs:722-740`].

[PROPOSE] Isolates stay. Their founding reason is **VA identity** — NVIDIA, and UVM especially, treats host VA and guest VA as one address, so each guest process needs its own host address space. Forgetting this is what let bug #14 (identical-VA collision between two guest processes) back into the C. But under the single store they stop being *memory owners* and become **address-space containers**: `Publish`, `PublishVidmem`, `JoinFbLeaf`, `AliasFbLeaf`, `FbJoinPeek` and `ExportBacking` all delete, leaving `MapStoreSlice`, `AdoptVaSpace`, `channel birth`, `doorbell` and `control`. □ **[UNVERIFIED]** Route K already births channels in the *scratchpad's* process — if that holds for UVM's channels too, VA identity has already moved and this conclusion changes.

2.2 The sandbox — two stages, forced order

Stage A, taken by the parent at `clone` [CODE `spawn_unsafe.rs:1009`]: `CLONE_NEWUSER|NEWPID|NEWNET|NEWIPC|NEWUTS|NEWNS`, `raw SYS_clone`, then rootless uid/gid maps (`setgroups=deny` first, as the kernel requires). `execveat(AT_EMPTY_PATH)` on a **sealed memfd**, so the image cannot be substituted. `PR_SET_NO_NEW_PRIVS` before `exec`.

Stage B, taken by the child in its own `main`, before any worker thread exists [CODE `sandbox_unsafe.rs:770-891`] — and **the order is the security property**, which is why it is one call and not three: nested user `ns` → `MS_REC|MS_PRIVATE` / → 256 KiB `tmpfs` `root` (`nosuid,noexec`) → bind the two device nodes → `pivot_root(".", ".")` + `umount2(MNT_DETACH)` → **only now** open the `/dev` `dirfd` → remount `root` read-only → drop every capability and **read the result back**, a single surviving bit being a refusal.

□ **Three things any security claim must not overstate:** 1. **seccomp is NOT installed** [CODE `lib.rs:57`]. The fixed fd numbers exist *so that* a future filter can hardcode them; that filter does not exist. Any document saying the isolate is `syscall-filtered` is wrong. 2. **The CUDA scratchpad has a full filesystem view** for the duration of its CUDA bring-up, which runs *before* `sandbox::enter` [CODE `child.rs:355-374`]. Constraint 20 must be read as “*it ends with the same reach*”, not “*it never had more*”. 3. That bring-up makes the child **multi-threaded** before `sandbox::enter`, and `unshare(CLONE_NEWUSER)` is refused to a multi-threaded process — so the scratchpad depends on the fallback arm under inherited `CAP_SYS_ADMIN`.

□ [CODE `isolate.rs:20-32`] fd 4 is **deliberately vacant**. It used to carry an `0_PATH /dev` `dirfd`, and `openat(4, "../etc/shadow")` **opened** — `0_PATH` places no restriction on ...

2.3 The threads

Exactly two production threads are created by `kayfabe` inside the VMM [CODE]:

thread	loop	waits on
<code>kayfabe-doorbell-publish</code>	<code>doorbell_publish_loop</code>	<code>Condvar</code> on the publication queue
<code>kayfabe-completion-observer</code>	<code>observer_loop</code>	<code>epoll_wait</code> , 250 ms tick, woken early by an eventfd

Plus QEMU's own `main` and `vCPU` threads, on which our trap code runs.

Inside each isolate child: a `main` thread, one **control** thread blocked in `recv(2)` on a datagram socket, and `--workers` (default **4**) pool threads each blocked reading its own socket. □ **None of the child's threads are named**, so `/proc/<pid>/task/*/comm` shows `kayfabe-isolate` for all of them.

□ The control thread is separate for one reason: a cancel must be serviced **while every worker is busy**, which is the only state in which a cancel is ever wanted. It signals with `tgkill(SIGUSR1)` whose handler is installed with **no SA_RESTART** — “the single most consequential line in the file” [CODE `signal_unsafe.rs:63`] — because that is what makes an in-flight RM `ioctl` return `EINTR`.

□ **[UNVERIFIED]** `observer_loop` declares no `ThreadClass`, so it defaults to `Coordinator`, under which long blocking sections are a reported violation — yet it deliberately performs millisecond-scale budgeted drains. The publication worker *does* declare `Worker`. There is no comment either way.

3. The vCPU trap path, step by step

3.1 How a guest access reaches Rust

```
KVM vmexit
→ nvkvm_trap_write()          qemu/hw/misc/nvkvm/nvkvm.c:591  (MemoryRegionOps.write)
→ kayfabe_shim_regs_write()    shim_unsafe.rs:1371    (extern "C")
    mark_vcpu_thread()          :1384                thread-local
    TrapGuard::enter_at(bar<<56|off) :1453            thread-local depth
→ Regs::write()                shim.rs:18576
→ RegPlane::write()             plane.rs:6473
```

Reads are shorter: `nvkvm_trap_read` → `kayfabe_shim_regs_read` → `Regs::read` → `RegPlane::read`.

∅ **BAR0 is cut into pieces**: 4 KiB pages whose every dword is “dead” get a RAM-backed alias so the guest’s reads of them never exit at all; the rest stay live trap pieces.

3.2 A DOORBELL write (BAR0 0x00BB0090) — the hot path

Defaults in force: `doorbell_async = 0n (defers)`, `doorbell_inline = 0ff`.

1. `RegPlane::write` — two atomics (writes, `mmio_in_flight AcqRel`), then the doorbell arm matches and **returns before the rank-0 plane mutex is ever reached** [CODE `plane.rs:6525`].
2. `ring_doorbell` — bumps a counter; takes `mmu_inval.inner` briefly; takes **plane.doorbell (read)** across the call.
3. `SharedDoorbell::ring` — the passthrough inline short-circuit is off by default; `shadow_route` is pure arch math + one Relaxed load on an atomic slot array. **No lock**.
4. `pubqueue.offer(MapPublication::for_doorbell(token))` — takes **pubqueue.inner**, coalesce check, cap check, `VecDeque::push_back`, **drops the guard**, then `notify_one`.
5. Back out: `dbcost (atomics)`, `dbledger.seen` (a mutex, ≤64 tokens), and the report is `Scheduled` — so `announce_completion` is skipped and `state/cpu_intr` are **not** taken.
6. The tail of `Regs::write` runs on every write: a few Relaxed epoch swaps, each of which may emit one further offer.

Net on the vCPU: ~6 atomics, **4 short lock acquisitions**, one `VecDeque` push, one `notify_one`. No rank-0 plane lock, no device lock, no IPC, no blocking.

Deferred to the worker: everything — mirror revalidation, isolate spawn, ring adoption, the GPFIFO read, the host forward or CE execution, the retired-object drain.

3.3 An MMU_INVALIDATE write (BAR0 0x00B830B0) — the one the guest spin-polls

1. `RegPlane::write` → the invalidate arm. Writes to the two PDB latches are pure latching.
2. On the **trigger**: `note_trigger` takes `mmu_inval.inner`, decodes, then `pending.swap(true, AcqRel)` — *this is the atomic the guest polls* — and bumps `issued`.
3. `shadow_write` pushes the new trigger value into the BAR0 backing page: takes **plane.read_shadow (read)** then `ShadowSink::segments`, then a `write_volatile`.
4. `Regs::write` offers `MapPublication::for_invalidate` onto the same queue.
5. **The vCPU returns. It never blocks.**

Where the guest blocks: in its own loop. `ogkm` spin-polls the trigger until it reads false, with a **4 s** (graphics) / **30 s** (compute) timeout. If the containing page is RAM-backed the poll costs **no vmexit at all**; if not, each poll is a read trap served by a deliberately lock-free arm. □ **[UNVERIFIED]** whether page `0xB83000` is backed for the shipped profile — the predicate is per-dword over the whole page and the survey did not execute it.

What clears TRIGGER: the worker, at the end of its pass — `complete_through_unmaps(seq, ...)`, which **withholds** completion if a newer trigger arrived or if unmaps are outstanding (Constraint 27: completing early would let the guest reuse a page still reachable through a slice we have not torn down), then `pending.swap(false)` and a `write_volatile` of 0 into the shadow.

□ [CODE shim.rs:18726] the completion is **no longer a Drop guard**. In the code’s own words: *“complete() is now the ONLY thing that clears the trigger, and a worker that dies without calling it hangs the guest by construction.”*

3.4 Back-pressure — the rule that makes deferral safe

The queue never drops work silently. A refused offer converts a **notification** into a conservative **full sweep**: a dropped doorbell arms an emulated-channel sweep; a dropped invalidate arms a full rescan; a dropped RPC bind force-releases held replies. Owner’s rule, quoted in-source: *“a lost NOTIFICATION becomes a conservative FULL RESCAN ... losing one may only ever cost time, never coverage.”*

□ **Coalescing is OFF for doorbells, and that is a measured verdict.** [MEASURED] on real GA106 the coalescing arm produced cuCtxCreate → unknown error (999) and two host Xid 31S (CE2 FAULT_PDE, GR0_PBDMA0 FAULT_PTE); the nocoalesce arm was clean. *“Deferring is fine and coalescing is not.”*

4. Locks, and the blocking rules

4.1 The ranked order

[CODE lock.rs:56] *“The total, one-way lock order ... Every lock declares its rank at construction; a thread may acquire only in strictly increasing rank, at most one lock per rank.”*

rank	lock	protects
0	RegPlane::state	the GSP FSM, guest-RAM port, command policy
1	RegPlane::mem	the emulated framebuffer store + GMMU format
2	SharedDevice::state (RwLock)	the spine, routing maps, delivery pump
3	per-Proc mutex	one guest process’s channels/VA spaces
4	leaf (cpu_intr, executor inbox, recorder)	

Enforcement is real: check_acquire **panics before the OS acquire**, always on, not a debug_assert.

□ Why the plane sorts *below* the device despite being further out: the shipping order is plane→core and ranks must strictly increase, so the plane must sort first *“or every vCPU MMIO trap in the device would panic.”* The ABBA this fixed was **guest-buildable** — *“A guest could build the deadlock by ringing a doorbell on one vCPU while touching a register on another.”*

4.2 The blocking rules — what is actually enforced

rule	mechanism	teeth
R1 no blocking under a ranked lock	assert_lock_free	runtime panic, always on
R0 no blocking on a vCPU	assert_not_on_vcpu	□ census by default — panics only under KAYFABE_VCPU_BLOCK_FATAL
no publishing from a trap	OffVcpu token, taken by value	□ compile-time ; minted in exactly one place
INLINE-SAFE(site) predicate	prose only	∅ (a) and (b) have no mechanism anywhere

□ The OffVcpu doc is the clearest statement of the principle in the tree: *“A constraint that lives only in prose is re-litigated by every new path, and the reviewer who has to notice it is the weakest link in the system. **The type is the reviewer.**”*

□ **The read path does not mark the thread as a vCPU.** kayfabe_shim_regs_read installs only the TrapGuard; a thread that has never done an MMIO write is not is_vcpu(). A second, independent detector does catch it — *“the trap witness is measured, a declaration is a claim, and where they disagree the measurement wins.”*

4.3 The gate that is RED right now

tests/tests/unranked_locks.rs enumerates every **unranked** lock a vCPU can hold, because `assert_lock_free` masks only *ranked* locks and therefore **cannot see these at all**. Its own header is honest about what it buys: “*ENUMERATED, NOT ENFORCED ... Nothing fires when one is taken in a bad order.*”

[MEASURED 2026-09-20] it FAILS with **21 undeclared** unranked vCPU-path locks. Two matter most:

- `dbtable.rs Mutex<BTreeMap<u64,[u64;4]>>` — **the doorbell ledger, taken on every doorbell**. ∅ Mine, added w801.
- `shim_unsafe.rs Mutex<Vec<ShadowSegment>>` — taken by the **vCPU** in `shadow_write` *and* by the **worker** in `publish_invalidate_trigger`. That is a lock genuinely shared between a trap and a worker, which is **precisely clause (c)** of the inline-safe predicate.

□ And if the BQL contradiction in §0 resolves toward *lockless*, two vCPUs can be in these paths simultaneously, which makes the shared-lock case sharper still.

[PROPOSE] Classify all 21 with real rulings — not to turn the gate green, but because the answer to “*can anything block beneath this?*” is a fact the v2 design needs anyway.

4.4 Where we block on purpose

[CODE `isolate.rs:454`], stated at the site: “*this is a synchronous write-then-blocking-read over a unix socket to the isolate child, and every production caller is inside the vCPU’s MMIO trap under the QEMU BQL ⇒ whatever the child spends here, the guest pays.*”

⇒ So the slogan “no blocking work in any MMIO trap” is **not** what the code does. There is one sanctioned blocking site and it is the isolate round trip. The store-map path explicitly refuses to be one of these, declining by name on a vCPU rather than stalling.

[MEASURED w470] the worst recorded stall was an isolate **spawn** on a vCPU — `Regs::write → materialize_pending → spawn_host → read_frame → __recv`, **1.62 s** inside an MMIO exit. That is why the scratchpad is brought up at realize, before the guest runs.

[PROPOSE] v2 should make this structural rather than sanctioned: no RM verb on a vCPU at all. Every such call becomes an offer onto the queue, and the guest waits on its own primitive — which §1 of the architecture says it already tolerates.

5. Data structures

5.1 The object graph — identity vs label

The whole file is built on one distinction, and it is the right one: **a guest handle is a label, not an identity**. RM recycles handle values with no quarantine and no free list, so the same 32-bit number names different objects over time.

type	what it is
<code>NodeKey { client, handle }</code>	a handle — per-client namespace slot. A label.
<code>ResId(u64)</code>	stable identity, minted at the origin alloc, never reused
<code>ResourceKey { origin, incarnation }</code>	the reportable identity; ordered origin-first so maps iterate deterministically
<code>HandleRef::{Origin, Alias}</code>	how a live handle came to reference its resource — recorded, never re-derived

```
struct Resource {  
    node: RmNode, refs: BTreeSet<NodeKey>, pdb: Option<Pdb>,  
    pdb_aperture: Option<Aperture>, gpu: Option<GpuId>, map_refs: usize,  
    owner_kind: ClientKind, owner: ClientId, owner_incarnation: u32,  
}
```

Liveness is `refs` non-empty **or** `map_refs > 0`. `next_incarnation` is “*the smallest ordinal not held by a live resource at this handle value*”, computed as a **range query**, not a scan — explicitly to deny a complexity DoS.

∅ `RmGraph::apply` **clones the whole graph** for rollback, because three faults are raised after mutation. The comment records it: ~24 % of an O(live objects)-per-event control plane. **[PROPOSE]** this is a named, deferred cost and v2 should fix it by validating before mutating.

5.2 The runtime model

```
Gpu { spine: Spine, system: Proc, procs: BTreeMap<ProcId, Proc> }
```

□ The concurrency contract is in the type, not a comment: a **per-proc** op takes the device *read* lock plus that proc’s mutex; a **spine** op takes the device *write* lock plus exclusive access to every proc, expressed as a ProcSet argument.

Proc is not a hardware concept. ProcId is “purely the label of one dup-connected component of declared user clients in the RM graph.” The stable name is ProcAnchor — the smallest client *declaration* in the component.

□□□ The distinction that earned its keep: clients (ClientKey) **labels** a component; client_ids (ClientId) **matches** one. Matching on handle values alone let a re-declared namespace **adopt the previous tenant’s proc** — its isolate, arena, host VAS and pending-release queue.

Vas carries the address plane; Channel carries the exec plane. $\emptyset$ Channel::method_state is the one guest-driven state and is bounded **structurally** (a fixed array), not by a check.

5.3 The address table — what v2 deletes

```
pub struct AddressTable { map: IntervalMap<Row>, owner: Option<Pdb>, generation: u64, blockage }  
pub struct Binding { kind: RegionKind, phys: u64, aperture: Aperture, host: Option<HostBacking> }
```

All four Binding fields are private with exactly two constructors — declared_by_guest (no host object) and real_gpu_memory (the only kind carrying one). Rows are **ranges**, not pages.

resolve(pdb, va) checks identity **first** (owns), then a predecessor lookup; a miss is a **fault**, never a heuristic. bind checks identity, then host-VA identity, then extent agreement, *before* anything enters the map.

[PROPOSE] This whole structure goes. Under v2 the walk produces a diff, the diff is mmap/munmap, and the **host page tables are the model**. Every field above is already in hand at the moment you act: the target is identity (store offset == FB offset), system comes from a static layout, permissions come from the PTE just walked, and “it went away” is munmap.

□ The machinery is **already built and unwired**: kayfabe_mmu::walkdiff::diff exists fully implemented with **zero production callers** (its only caller is a #[test]), MapOp::{Map, Unmap, Remap} exists, and StoreMapPort::apply_ops takes a **list** and does unmaps-before-maps — while its one production caller always passes a single-element Map. The source even says “when the delta lands, this call site does not change.”

$\emptyset$ **Consequence today:** the only production unmap is a same-VA re-point. release_vas has zero callers. So StoreMapPort::outstanding grows **monotonically over the VM’s life** — mappings only ever accumulate, which is the other half of the cross-client leak.

5.4 Crate layering

util → arch → abi → {mmu, core, fwd, rmrpc, device, rt} → qemu-raw, with vmm and isolate as ports.

□ One layering fact worth arguing about: **kayfabe-mmu depends on kayfabe-isolate**, because Binding carries a HostHandle. So the address plane sits *above* the sandbox port rather than beside it. **[PROPOSE]** deleting the address table inverts that and makes kayfabe-mmu a leaf crate — which is a good structural signal that the deletion is the right shape.

5.5 The isolate boundary — two layers, not one

□ A correction to a premise I held: VerbPlan/VerbReply are the **in-process** typed plan and reply. The **IPC** is Request/Reply/Envelope.

- **Transport:** one SOCK_STREAM socketpair **per pool worker**, granted at fixed fd numbers.
- **Framing:** u32 LE length then body, in **one** write_all — because a length written separately from its body permanently desynchronises a signal-interrupted channel.
- **No multiplexing, no demux, no txn correlation.** Each worker owns its socket and the channel is 1-deep. An explicit divergence from the C, which multiplexed 16 workers over one fd.
- $\emptyset$ Two RmError variants deliberately have **no wire form**: ForeignHandle (our own gate produces it before a request is written — a compromised child must not be able to claim our invariant broke) and Wedged (a child that could send it would be reporting that it never answered).
- **Cancel** travels out-of-band on a separate datagram socket, guarded by a txn the worker publishes immediately before the verb and clears immediately after — without it a cancel races the completion and lands on an unrelated later operation.

6. Installing a mapping, and observing a completion

6.1 The mapping chain, end to end

The decision is made by a **publication pass**, not by a fault — we map ahead, we do not wait to be asked.

```
doorbell_publish_loop          (worker thread; mints the one OffVcpu)
→ publish_vas_rows(token, seen, OffVcpu) shim.rs:13259  - takes OffVcpu BY VALUE
→ vas_publish_census(...)      - produces candidate (va, len, phys)
→ join_one_fb_leaf(...)        shim.rs:15099
→ plane.fb_join_plan(phys, len) → FbJoinPlan::DeviceBacked { at: phys }
→ map_store_slice_for_leaf(...) shim.rs:14765
    step 0  decline if on a vCPU or in a trap          :14787
    step 1  vaspaces_handover → scratchpad dups the space (once per Vas)
    step 2  apply_ops(store_vas, [MapOp::Map{ va, gpgas: at, len }])
→ StoreMapPort::map          storemap.rs:559
→ iso.with_worker(|w| w.with_rm(|rm| rm.map_store_slice(...)))
→ ProxyRmBackend::map_store_slice → write_frame / blocking read_frame (process boundary)
→ child: worker_loop → serve_one → HostRmBackend::map_store_slice
→ RmConnection::raw_map_dma_slice      rm.rs:4425
→ ioctl(NV_ESC_RM_MAP_MEMORY_DMA)      rm.rs:4509      ← THE SYSCALL
```

□ **The syscall happens in the scratchpad isolate child.** The VMM performs no RM ioctl for a store slice; it writes and reads one socket frame.

Two details that are load-bearing and easy to lose: - ∅ The code takes **RM’s answer, not its own request** (`done.placed.first()`). The comment records that substituting the requested address here regressed the guest to NEVER RETIRED. - ∅ **Constraint 28:** if RM places the mapping anywhere other than where we asked, the mapping is **turned down** and the call refuses — a fixed map that silently moved is worse than none. - ∅ `status == 0x51` on a FIXED map means *already mapped there* (occupancy, not capacity) and is re-reported as such.

Guest RAM takes a different route: the memfd arrives **at spawn on a fixed fd**, not on a request, so a future seccomp filter can hardcode it. The child `mmaps` it and wraps it in an `OS_DESCRIPTOR`; ∅ RM pins those pages and releasing the `mmap` does **not** unpin them.

6.2 The single store

[CODE] Allocated at **PCI realize**, in the scratchpad, by a **bisecting probe**: halve to a floor, then bisect to 64 MiB grain, each attempt allocated and freed. First try contiguous + 1 GiB aligned; fall back to non-contiguous + 4 KiB, latching which — and that latch is what later picks the page-size flag for every mapping.

[MEASURED] **11904 MiB**. □ Not a constant anywhere: the compiled advertisement is 12288 MiB and is **rebound to whatever the probe reserved**. □ The identity guest FB offset == store offset is only safe *because* of that rebinding.

∅ Deliberately **not** `alloc_vdmem`: that verb is contiguous-with-alignment = len, so an 11.8 GiB request refuses on **fragmentation** — a refusal that is not about capacity, on the one call whose refusal is supposed to mean “there is not enough video memory.”

6.3 Completions — two mechanisms with opposite writers

path	who writes the value the guest polls	how we learn
GR / compute , forwarded	the GPU	we only <i>read</i> : <code>epoll_wait</code> at a 250 ms floor, woken early by an eventfd; the watch list is declared on the vCPU and swept on the observer thread
emulated CE , served locally	we do	our executor writes the guest’s own literal payload, at the address the guest’s own page tables resolve to, after the bytes moved

□ The completion-watch module **never writes, never raises, never resolves** — the capability it is handed is a reader, by construction. □ The CE writer is a **single-writer discipline** with a compile-fail test and a workspace-wide census, because the C had a second, lagging writer that corrupted a live payload.

∅ `AWAKEN_ENABLE = 0` in the measured `cuCtxCreate` stream: the guest asked for **no interrupt at all**. The completion is a value appearing at an address. That is why polling — not interrupt delivery — is the mechanism here.

6.4 Interrupts — what gets announced

Two independent gates, both in `ring_doorbell`: - **non-stall vector**: only a **locally served** doorbell announces. A forwarded one does not — the work finished on a host engine at an instant this device was not standing at. - **OS-event wake**: fires for **served or served-locally**. [MEASURED w686] gating this on locally-served alone meant **40 of 44** completions on passthrough channels announced nothing and the guest sat in `MC_SERVICE_INTERRUPTS`.

Ø `LEAF_TRIGGER` is the **guest's own** self-trigger register, not our announcement path. Masking is **recorded, not acted on**.

7. The proposal, and the arguments against it

This is the part to attack. Each item has the counter-argument I can see.

P1. Mirror, not adopt

There are two ways to make the GPU honour the guest's translations.

- **Adopt** — point the host channel's instance block at the **guest's own** page-directory root. No walk, no diff, no syscalls at all. Most of v2 evaporates.
- **Mirror** — walk the guest's tables, replicate into a host VA space.

I propose mirror, and I want this one argued. My objection to adopt: the guest's PTEs contain **guest** physical addresses, which are host-valid only where the backing is identity — true for the store, false for system GPAs. □ If adopt can be rescued (e.g. by making system identity too), it is strictly better and I would drop mirror.

P2. No address table; the host page tables are the model

Already argued in §5.3. The strongest evidence it is right: **the diff machinery already exists and is unwired**, and the reconciliation bugs that dominated 2026-09-19/20 are all of the form *"two copies of one fact disagreed."*

Ø **Argument against**: syscall volume. N changed pages $\Rightarrow$ N mappings unless runs are coalesced. **Unmeasured**. The walk is 205.7 μ s and page tables change far less often than doorbells ring, so I expect small bursty diffs — but that is a prediction.

P3. Lazy everywhere; strict only at recycle — and recycle is the guest's job

A doorbell **never** waits for a refresh. `ogkm` already accepts that addresses are *undefined* until an invalidate completes: old page or new page, never a page neither mapping referenced.

$\Rightarrow$ We own no scrub policy. Our single failure is that we **forge the scrub's completion** (`forwarded=0` on the kernel tokens), so the guest reuses a page it believes was cleared. **That is the cross-client leak**. Fix: let the work reach the GPU. The gate already exists.

P4. No RM verb on a vCPU, structurally

§4.4 shows the current design **sanctions** one blocking site and pays for it with a measured 1.62 s stall. **[PROPOSE]** make it impossible instead: every RM verb becomes an offer, and the guest waits on its own primitive.

Ø **Argument against**: some verbs are genuinely synchronous in the guest's eyes — a control command whose reply the guest reads immediately. Those need an answer *now*. □ I do not have a clean answer for that case and it is the main hole in P4.

P5. Deletion order

1. **The scrub** — live security bug, gate exists, smallest change.
2. **Pdb-as-identity** — still costing diagnoses; split into `VasId + Option<Pdb>`.
3. **The address table and joins together** — they are one mechanism.
4. **The CPU executor last** — the scrub result tells us which half of that plane survives.

P6. What must NOT be deleted

The instruments. The doorbell ledger's per-token `forwarded=`, the censuses, refusals **by name**, the bare-metal baseline, the hardware gate. [MEASURED 2026-09-20] the bare-metal baseline is what turned nine guest failures from opinion into adjudication; without it, a rewrite cannot be graded, and a rewrite you cannot grade is a rewrite you cannot land.

8. Open problems — named, not designed away

1. **Where the walk's root comes from.** `MAP_MEMORY_DMA` is a HAL stub on GSP parts, so a client-allocated VA space is never declared — [MEASURED] **zero** `SET_PAGE_DIRECTORY` events in a whole boot. Candidates, ranked: the **instance block** (which we author, and which is how hardware itself finds the root); `COPY_SERVER_RESERVED_PDES` (arrives, but only for RM-managed spaces); observing the guest's own page-table writes.
2. **Fault delivery becomes load-bearing.** v2 lets the GPU fault instead of pre-validating operands — but guest fault-recovery methods are currently **refused by name** precisely because we do not deliver faults. That refusal is the marker for this work.
3. **The BQL question** (§0). It changes the concurrency argument wholesale.
4. **The 21 unranked vCPU-path locks** (§4.3).
5. **Multi-GPU and non-GA10x.** The chip table holds exactly one profile.
6. **Syscall cost of the diff** (P2).

Part 4 — The surface we present (v2 plan-level)

STATUS: LIVE, 2026-09-20 (w815). Companion to `THE_ARCHITECTURE_v2.md`.

∅ Every row below was read out of the tree at w749-fable-legb, with file:line. Where the survey could not establish something it says so — **an unverified row is marked □, never filled in**. This document is the inventory the v2 rewrite is scoped against: the **plan** column is the decision, and `[DELETE]` means the mechanism goes away, not that the behaviour does.

1. GSP RPCs

The wire table is generated from `rpc_global_enums.h` (kayfabe-abi/src/generated/rpc.rs); classification is `Function-Codes::classify` (kayfabe-gsp/src/rpc.rs:209); the one dispatch is `translate()` (kayfabe-rmipc/src/lib.rs:1073).

1.1 The object-model RPCs — KEEP, unchanged

GSP_RM_ALLOC (0x67), GSP_RM_CONTROL (0x4c), FREE (0x0a), DUP_OBJECT (0x15).

These four are the whole guest→us object protocol: the guest tells us what it allocated, what it freed, what it aliased, and what it wants done. We decode them into `RmEvents` and keep a graph of the guest's objects. **Plan: unchanged in v2.** This is the control plane, it is version-sensitive, it is heavily tested, and none of v2's deletions touch it. An unmapped alloc class is refused by name (`UnmappedAllocClass`) and that stays — a class we do not model is a thing we must not pretend to have created.

1.2 UPDATE_BAR_PDE (0x46) — KEEP, but it is the BAR2 root

Inert to the object model; acted on at `kayfabe-device/src/bar2.rs:299`, where it latches the root PDE that BAR2 translations are rooted at, and replies `NV_OK` with an empty body. **Plan: keep.** BAR2 is an instance-block window and the guest legitimately re-roots it. □ This is one of the few places v2 still walks page tables *to answer a read*, rather than to produce a diff — see §3.4.

1.3 SET_GUEST_SYSTEM_INFO (0x01) / _EXT (0x40) — KEEP

The version handshake. We answer with the version we agreed to speak; the `_EXT` form is answered `NV_OK` with an empty body. The guest's firmware string is latched read-only. **Plan: keep** — this is where the support matrix is enforced, and it must stay strict.

1.4 GET_GSP_STATIC_INFO (0x41) — KEEP, forged by construction

We synthesise an entire `GspStaticConfigInfo` from the chip and driver tables (`staticinfo.rs:183`). There is no host value to forward: on real hardware this comes from GSP firmware we are replacing. **Plan: keep forged.** □ This is the single largest fabricated structure we hand the guest and it is derived per-die; it is the first thing to re-derive when adding a chip.

1.5 The inert set — KEEP as inert, they are load-bearing

INIT_GSP_TRACE_CRASH_BUFFER (0xe4), UNLOADING_GUEST_DRIVER (0x2f), GSP_SET_SYSTEM_INFO (0x48), SET_REGISTRY (0x49), ECC_NOTIFIER_WRITE_ACK (0xca).

Accepted, no state moves. ∅ Two of these matter more than “inert” suggests: `UNLOADING_GUEST_DRIVER` **requires a reply** — a missing one hangs `rmmod` — and the two `NoReply` ones must **not** be answered, because an unsolicited reply desynchronises the queue. **Plan: keep, with the reply disposition as the thing under test.**

1.6 Events we post — KEEP

EVENT_GSP_INIT_DONE (0x1001), EVENT_POST_EVENT (0x1003), EVENT_RC_TRIGGERED (0x1004). We send these; a guest-sent one is refused `EventFromGuest`. **Plan: keep.** The refusal direction is a real boundary — the guest must not be able to inject its own completions.

1.7 Everything else — refused by name, and that is the design

Other(code) ⇒ UnknownFunction { code } — *replied to, never dropped*. Includes ids present in the generated table with no dispatch arm: NOP, ALLOC_ROOT, ALLOC_MEMORY, **MAP_MEMORY_DMA (0x0e)**, UNMAP_MEMORY_DMA (0x0f), EVENT_GSP_RUN_CPU_SEQUENCER.

□□□ **MAP_MEMORY_DMA is the important one**. It is a HAL stub on GSP-client parts, so the guest never sends it — which is why [measured w811] a whole boot carries **zero** SET_PAGE_DIRECTORY events and a client-allocated VA space is never declared to us. **Plan: this is open problem #3 in THE_ARCHITECTURE_v2.md §7**. v2 needs a root to seed a walk and this RPC will not provide it.

2. RM control commands

2.1 Page-directory controls — [DELETE the concept, keep the decode]

cmd	name	today
0x00801813	DMA_SET_PAGE_DIRECTORY	→ RmEvent::SetPageDir
0x90f10106	VASPACE_COPY_SERVER_RESERVED_PDES	→ RmEvent::SetPageDir
0x20800a9f	GMMU_COPY_..._PDES_TO_SERVER	same params as above
0x00801814	DMA_UNSET_PAGE_DIRECTORY	refused PageDirControlNotModelled

These tell us a VA space’s page-directory base. **Plan: keep decoding them, but demote what the answer is used for**. In v2 a root is needed **only to seed a walk** — never as an address space’s identity. [ROT] today Pdb is both, and unwrap_or(Pdb(0)) at eight sites makes Pdb(0) mean “declared at offset 0” and “not declared”; that collision cost four wrong diagnoses in one session. □ Note both arms refuse a **sysmem** root today (SetPageDirRootAperture / PublishedPdesRootAperture) even though a real GA106 was measured with one — that refusal is on the v2 review list.

2.2 GPU_PROMOTE_CTX (0x2080012b) — KEEP

The guest tells us where it placed a context buffer. Decoded, ≤16 entries, entryCount > 16 refused. **Plan: keep**. □ The C rounded every promote-derived mapping **up to 64 KiB** (asize = (size + 0xffff) & ~0xfffffull); this port binds at the declared length and that difference produced sub-page holes. Under v2 the mapping comes from the **walk**, not from the promote, so this becomes an observation rather than a mapping source — which dissolves the rounding question instead of answering it.

2.3 Channel controls served by ObjectPolicy — KEEP

GPFFIFO_SCHEDULE (0xa06f0103 / 0xa06c0101), BIND (0xa06f0104), SET_CTXSW_PREEMPTION_MODE (0x20801210), MC_SERVICE_INTERRUPTS (0x20801702), GET_MMU_FAULT_INFO (0x906f0106), PREEMPT (0xa06c0105).

The channel lifecycle: schedule it, bind it to an engine, preempt it, ask why it faulted. **Plan: keep all**. ∅ GET_MMU_FAULT_INFO is **relied to the host channel verbatim, never synthesised**, and that becomes *more* important in v2: if the GPU faults because our diff was late, the guest must receive the hardware’s own answer.

2.4 The 47 forged tables — KEEP forged, but they are a per-die liability

FIFO_GET_DEVICE_INFO_TABLE, INTERNAL_GPU_GET_CHIP_INFO, CE_GET_CE_PCE_MASK, FB_GET_INFO_V2, GR static caps, the cudart-init group, and ~40 more (kayfabe-device/src/inittables.rs:1085). Each is answered with a struct built from our chip and driver tables; a paramsSize mismatch is refused.

Plan: keep, and derive per die rather than maintain per die. □ This is the largest correctness surface we own and the one with the worst oracle: [the C artifact] 11 of 56 captured rows had **dlen = 0** and every empty row checked against real hardware was **contradicted**. CE_GET_FAULT_METHOD_BUFFER_SIZE decoded from an empty row as **0** where a real GA106 returns **20480** — and RM DMAs fault records into a buffer of that size, so trusting it was a buffer overrun with a hardware writer.

2.5 Input-only controls — KEEP

NV2081_BINAPI, DEBUG_SET_EXCEPTION_MASK, SET_TIMESLICE, two CUDA-limit controls. The ack is the whole verb; we echo the guest’s own bytes behind a size check. **Plan: keep**.

2.6 Denied by capability — KEEP DENIED

IMPORT_MEM, DISABLE_IMPORTERS, NVLINK_GET_PLATFORM_INFO (fabric management); GPU_EXEC_REG_OPS, NVB0CC_EXEC_REG_OPS (raw register access); REPORT_NON_REPLAYABLE_FAULT; the SM-debugger trapping set; ALLOC_PMA_STREAM (performance counters).

Plan: keep denied, permanently. These are the guest asking for a capability the host process does not have and must not acquire. □ This is a *capability allowlist* (160 rows), not a served set: admitted-but-undecoded still refuses GspRuleControlUnserviced. □ The survey did not transcribe all 160 rows — that is a mechanical dump if the full admitted set is ever needed.

3. MMIO registers

□ **GA106 only.** The chip table holds exactly one profile; ad10x, gh100, gb20x exist in kayfabe-chips and are not in it. □ The BAR0 aperture is 16 MiB and ~87% of its pages hold no register at all — “everything else = Unclaimed” is the complete statement for the remainder.

3.1 Boot constants — KEEP

NV_PMC_BOOT_0/1/42, NV_PTIMER_TIME_PRIV_LEVEL_MASK, NV_USABLE_FB_SIZE_IN_MB, NV_XVE_LINK_CAPABILITIES. Constants the driver reads to identify the part. **Plan: keep, per-die.** ∅ One is an advertised fiction — the access-counter notify buffer size carries an entry count only.

3.2 The GSP boot FSM — KEEP

GFW_BOOT_PROGRESS/PLM, the PGSP/PSEC falcon registers (CPUCTL, HWCFG2, DMATRFCMD, MAILBOX0/1, IRQS-TAT/MASK/DEST/SCLR), NV_PRISCV_*, WPR2_ADDR_LO/HI.

This is the fake GSP’s entire visible boot: the driver pokes STARTCPU, watches progress reach 0xFF, reads mailboxes for boot args, and finds WPR2 where it expects. **Plan: keep.** ∅ WPR2 state only resets on a full QEMU restart, which is why each clean run needs a fresh boot.

3.3 Queue doorbells and the interrupt tree — KEEP

NV_PGSP_QUEUE_HEAD(i) (0x110C00 + i*8) is the command doorbell: posted to a worker, no state touched inline. CPU_INTR_LEAF(i) / EN_SET / EN_CLEAR, CPU_INTR_TOP*, and CPU_INTR_LEAF_TRIGGER (write-only, latches a vector and raises the CPU interrupt).

Plan: keep. ∅ [measured w684] 40 of 44 completions were never announced (no_engine); the interrupt **arming** model is the live design question here, not the register decode.

3.4 The MMU invalidate trio — [DELETE most of what it triggers]

MMU_INVALIDATE_PDB (0xB830A0) and _UPPER_PDB latch; MMU_INVALIDATE (0xB830B0) **triggers** a publication, and the guest **spin-polls** its TRIGGER bit for completion.

Plan: keep the register, delete the publication behind it. In v2 an invalidate is a hint that the guest’s tables changed — a good moment to walk and diff — and **nothing waits on it** (§1: the guest already tolerates undefined-until-complete). ∅ Today this register is the entry point to the epoch/dirty-gate/publication machinery that §3 of the architecture deletes. □ RM cannot express a narrow invalidate (ALL_VA is hard-coded) and its own timeout is **4 s**, after which it proceeds with stale TLBs — so the guest is *already* designed for us to be slow here.

3.5 The framebuffer windows — [DELETE]

NV_PBUS_BAR0_WINDOW (0x1700) plus the PRAMIN moving window (0x700000–0x7FFFFFFF); BAR1 (bus_bar::FB) and BAR2 (bus_bar::INST), both GMMU-translated on access; FbTrapPolicy.

Plan: BAR1/BAR2 become device views of the one store and stop being translated windows; PRAMIN stays as the bring-up aperture it is, not a running path. ∅ [measured] vidmem CPU reads are ~48 MiB/s flat — any design that routes real traffic through these is slow by construction, which is precisely why the store exists.

3.6 The doorbell — KEEP, it is the hot path

NV_VIRTUAL_FUNCTION_DOORBELL (BAR0 0x00BB0090). Rung off-lock; drives channel submission. NV_VIRTUAL_FUNCTION_TIME_0/1 beside it are the PTIMER: read-only, **writes refused by name** (the guest reads the host’s counter and may not move it). **Plan: keep all three exactly.** ∅ §41 bounds what this trap may do: update a queue, wake a worker, one dword for a passthrough doorbell, or a synchronous read-register write. Nothing else, ever.

4. Channel / pushbuffer operations

Decoder: Ga10xPushbuffer (kayfabe-chips/src/ga10x.rs:1595); vocabulary PushMethod (kayfabe-arch/src/lib.rs:949); consumer apply_pushbuffer (kayfabe-fwd/src/lib.rs:8721).

4.1 EMULATED — we implement the effect

method	addr	why it is ours
SET_OBJECT	0x0000	pure bookkeeping — binds a subchannel class, clears its slots
host-FIFO semaphore run (SEM_ADDR_LO/HI, SEM_PAYLOAD_LO/HI, SEM_EXECUTE)	0x5C–0x6C	RELEASE only ; acquires and reduction decode Opaque
MEM_OP_A..D TLB invalidate	0x28...	recorded and censused; PDB_ALL and non-invalidate ops = Opaque
LAUNCH_DMA with DATA_TRANSFER_TYPE_NONE	0x0300	a release carrying no copy — there is no transfer to forward

Plan: keep all four emulated. Each is either bookkeeping or a completion, never a byte movement. ∅ In v2 the TLB-invalidate record becomes a *walk trigger* and loses its publication side effect (§3.4).

4.2 TRANSLATED — operands rewritten, the GPU moves the bytes

LAUNCH_DMA (0x0300) is the only true member, fed by fourteen latched operand slots: OFFSET_IN/OUT_UPPER/LOWER, LINE_LENGTH_IN, LINE_COUNT, SET_SRC/DST_PHYS_MODE, SET_REMAP_CONST_A/B, SET_REMAP_COMPONENTS, SET_SEMAPHORE_A/B/PAYLOAD/_UPPER.

Today we re-resolve *src/dst* through the issuing channel’s **address table** and partition into spans. **Plan: [DELETE the translation].** Under v2 the guest’s VA is the host’s VA (route K) and the store is identity-mapped, so there is nothing to rewrite — the GPU resolves the operands through the page tables we applied, and faults if we were wrong. ∅ This deletes TableOperands, the span partitioning, the operand gate, and CeExecutor::ours with it.

□ SET_SRC/DST_PHYS_MODE is the one that must **not** be waved through: PHYSICAL bypasses the MMU, and a physical-mode copy passed straight to hardware is an escape class. It stays inspected — that is exactly what a **Translated** channel is for, and why Passthrough is not allowed to carry it.

∅ SET_REMAP_COMPONENTS is **required, never defaulted** — an unlatched remap refuses the launch. Keep that; a defaulted remap silently writes the wrong bytes.

4.3 REFUSED — the fault tripwire

GP100_UVM_SW_FAULT_CANCEL_A/B/C (0x0104/0108/010C) and CLEAR_FAULTED_A/B (0x0110/0114) abort the whole parse with UvmFault-MethodWithoutFaultDelivery. SET_OBJECT/NO_OPERATION on the same class pass through as routine.

Plan: keep refusing — and revisit when v2 lands. These say the guest is doing fault *recovery*, which presumes a fault-delivery mechanism we do not implement. □ v2 deliberately lets the GPU fault (§3), so fault delivery moves from “not modelled” to “on the critical path”, and this refusal is the marker for that work.

4.4 PASSED THROUGH

Everything else = PushMethod::Opaque, counted, untouched. Non-Incrementing method forms (NON_INC, ONE_INC, IMMD_DATA) are deliberately not latched. **Plan: keep.** ∅ Parsing — not placement — is what is forbidden on a passthrough ring; this is the rule in code.

5. □ Where this inventory is knowingly incomplete

1. The 160-row capability allowlist is not transcribed (gates admission, not service).
2. Registers are **GA106 only**; three other chip modules exist and were not enumerated.
3. There is no enumerable table of *unhandled* BAR0 offsets — ~87% of pages are Unclaimed.
4. kayfabe-rt/src/ceutils.rs holds a **second** PushMethod consumer; it was confirmed to use the same enum but **not** audited for additional method addresses.

5. `kayfabe-isolate-host/src/rm.rs` *emits* CE pushbuffers using the same constants; classified as a producer, not audited as a decoder.
6. Roughly a dozen `kayfabe-device` modules were not read in depth; two of them gate on `RpcFunction::RmControl` and may reference control ids not listed here.

Part 5 — The scrub question (open owner ruling)

STATUS: OPEN — awaiting an owner ruling on §46 vs §12.26. Measured 2026-09-20 (w813) on a GA106 / 580.159.04 guest, branch w749-fable-legb. Nothing here has been implemented; the SystemDataPlane rule is untouched.

What is measured

The DOORBELL-LEDGER, per token, on a thin-guest run whose every ledger row verified:

token	class	emulated	forwarded	reading (the ledger's own rule)
0x00000003– 0x00000007	the guest's CE channels	1	5	reached hardware
0x00000009 0x00010001	guest kernel / system	2 74	2 0	reached hardware never reached hardware
0x00010004	kernel / system	66	0	never reached hardware

“forwarded=0 with emulated>0 means it never went to hardware; forwarded>0 means it did” — the ledger's own text.

⇒ **The guest's copy work runs on the GPU. The kernel's does not, and it is 140 doorbells.**

Why this is a decision and not a bug

The kernel tokens are forwarded=0 **by design today**. `kayfabe_fwd::FwdFault::SystemDataPlane` implements `l1_concurrency.md §12.26`:

*“The SYSTEM proc has no data plane ... Guest-kernel work that would need a backing — **the CeUtils scrub**, the GR golden capture — is **forged** to the system proc's completion queue, **never forwarded**, so the system proc never mints host memory.”*

□ §46 (owner, 2026-09-19) names that **exact** example and reverses it:

“okay no real work kernel channel except the no-op or the ones that we must stub, so the scrub and CE, is never executed on CPU. Its always executed on GPU under the single store” and, when I mis-paraphrased it: *“Ceutils scrub is not a no-op, it genuinely needs to clear memory. Same for kernel ce”*.

Two rules, one workload, opposite answers. The date is part of the citation, and §46 is the later one — but §12.26 is documented as a boundary, so the supersession should be **stated** rather than inferred by whoever next touches the code.

The argument each way

For §46 superseding (my reading): - §12.26's stated *reason* is “so the system proc never mints host memory”. Under the single store that premise is obsolete: the store is **one reserved device-local object minted once at realize**. A guest-RAM pin against it mints nothing; it names an offset in something that already exists. - The scrub is real work with an observable end state — memory that must read as cleared. §46 is explicit that this is not a no-op we may fake. - The same reasoning already moved the guest's CE copies onto hardware, and nothing about the system proc makes its bytes different bytes.

Against / what to be careful of: - §12.26 is about a **lifetime regime**, not a byte count: the system proc's isolate exists for the device's own bring-up and is not torn down with any guest process. Handing it data-plane state is a new lifetime question — *what frees these pins, and when?* - `ce_executor_tree.md` (owner, 2026-08-07) separately rules that `ce_copy(0urs)` must keep refusing inside the isolate, calling it “the security boundary refusing to leak guest memory into the sandbox, working as designed”. ∅ That is a **different** refusal from `SystemDataPlane` and is **not** what this page proposes changing — but the two are adjacent and must not be conflated in a hurry.

If the ruling is “§46 wins”, the work and its gate

1. `plan_pin_guest_ram` stops refusing `Gpu::SYSTEM_PROC` — and the lifetime question above gets an answer in the same change, not later.
2. The scrub's CE doorbell forwards like any other.

3. **The gate already exists and needs no new instrument:** tokens 0x00010001 and 0x00010004 must flip from forwarded=0 to forwarded>0 in the DOORBELL-LEDGER, and run_fast_guest.sh's stranded-token check must be widened from guest tokens to all tokens. □ Until then that check is deliberately scoped to tok=0x000000xx **precisely so it cannot pre-empt this ruling** — a gate that asserted the answer would make the question unaskable.

What must not be read into this page

∅ This is not a claim that the scrub is broken. It is forged, deliberately, by a rule that was correct when written. The question is only whether the rule survives the single store.